

Feldman–Hájek, but Non-Gaussian: Geometry of the Weighted Fourier Jet

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Abstract

Feldman–Hájek theory classifies equivalence of Gaussian measures; Kakutani’s theorem compares product measures. Neither criterion classifies general linear mixing of a non-Gaussian product, whose output coordinates need not remain independent. We give this classification for centered non-Gaussian i.i.d. coordinates with finite variance, a positive density, and finite location and scale Fisher information. A bounded invertible linear mixing preserves the measure class exactly when it is a Hilbert–Schmidt perturbation of a coordinate permutation with law-preserving signs; otherwise the image is singular. Thus the Hilbert–Schmidt scale of Gaussian equivalence survives, but its reference symmetries reduce from the orthogonal group to the coordinate symmetries of the product. The new step is to recover this operator criterion without a product factorization of the image law. The weighted Fourier jet detects non-Gaussian mixing through dimension-free Hellinger bounds, and tail exchanges turn local detection into square summability. For symmetric stable products, the off-diagonal cost becomes $\ell^\alpha(\ell^2)$, while scales retain a quadratic cost. For densities with zeros, forward and inverse preservation of the positivity set supplies the additional condition for equivalence.

1 Introduction

Which infinite-dimensional linear transformations preserve the null sets of a product measure? Feldman–Hájek theory answers this question for Gaussian laws through a condition on the covariance operators [6, 10, 12, 33]. Kakutani’s theorem answers it when both laws are products, by multiplying their coordinate affinities [17]. General linear mixing of a non-Gaussian product falls outside both criteria: Gaussianity is absent, and mixing destroys the independence needed for Kakutani’s factorization. The missing classification concerns exactly these transformations, which mix the coordinates without either classical structure to decide equivalence.

We give a necessary and sufficient operator criterion. For centered non-Gaussian i.i.d. coordinates with finite variance, a positive density, and finite location and scale Fisher information, a bounded invertible linear mixing preserves the measure class if and only if it differs from a coordinate permutation with law-preserving signs by a Hilbert–Schmidt operator. Every other mixing produces a singular law. The transformed coordinates may be dependent, and the operator may mix infinitely many inputs into infinitely many outputs.

The mathematical change from the Gaussian classification is the symmetry from which the perturbation is measured. A standard Gaussian product is invariant under every orthogonal mixing. For a non-Gaussian product, even preserving the measure class forces the mixing to approach the original coordinate axes, with square-summable error. The Hilbert–Schmidt scale survives the loss of Gaussianity; the freedom to rotate does not.

1.1 Main theorem

Write $\mu = \rho^{\otimes \mathbb{N}}$, $\rho(dx) = f(x) dx$, and $g = \sqrt{f}$. Our main assumptions are

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g \in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ X \sim \rho, \quad \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad f \neq (2\pi)^{-1/2} e^{-x^2/2}. \end{aligned} \quad (1)$$

Derivatives are weak. The two Sobolev conditions give finite location and scale Fisher information.

Put $H = \ell^2(\mathbb{N}; \mathbb{R})$, let $(e_i)_{i \geq 1}$ be its coordinate basis, and write $\mathcal{S}_2(H)$ for the Hilbert–Schmidt ideal [35]. Define

$$\begin{aligned} \mathcal{P} = \{Q : (Qx)_i = \varepsilon_i x_{\pi(i)}, \quad \pi \text{ a permutation of } \mathbb{N}, \\ \varepsilon_i \in \{-1, 1\}, \quad \text{Law}(\varepsilon_i X) = \rho \text{ for every } i\}. \end{aligned} \quad (2)$$

Permutations always preserve the product law; reflections do so precisely when ρ is symmetric. These are the symmetries collected in $\mathcal{P}$. For $T \in \mathcal{B}(H)$, write $\nu_T = \text{Law}((\sum_j T_{ij} X_j)_{i \geq 1})$, where the X_j are independent with law ρ . The rows belong to ℓ^2 , so each sum exists in L^2 . Their common measurable realization Φ_T is constructed in lemma 3.1.

To measure the accumulated change of law, we use Hellinger affinity and distance:

$$A(\lambda, \nu) = \int \sqrt{\frac{d\lambda}{dm} \frac{d\nu}{dm}} dm, \quad h(\lambda, \nu)^2 = 2 - 2A(\lambda, \nu). \quad (3)$$

Here m dominates both measures. With $\mu_n = \rho^{\otimes n}$ and π_n the projection onto the first n coordinates, the corresponding energy is

$$E_n(T) = -2 \log A(\mu_n, (\pi_n)_\# \nu_T). \quad (4)$$

Every input coordinate may contribute to these n outputs. Increasing n reveals more of the change of law and can only increase the energy.

Theorem 1.1 (Bounded invertible linear mixing). *Under (1), every $T \in \text{GL}(H)$ satisfies*

$$\boxed{\begin{aligned} \nu_T \sim \mu &\iff \exists Q \in \mathcal{P} : TQ^{-1} - I \in \mathcal{S}_2(H) \\ &\iff \sup_n E_n(T) < \infty. \end{aligned}} \quad (5)$$

If these conditions fail, then $\nu_T \perp \mu$ and $E_n(T) \uparrow \infty$. Moreover, $\nu_T = \mu$ if and only if $T \in \mathcal{P}$.

For the same operator T acting on a standard Gaussian sequence G , the Feldman–Hájek criterion reads

$$\begin{aligned} \text{Law}(TG) \sim \text{Law}(G) &\iff TT^* - I \in \mathcal{S}_2(H) \\ &\iff T - U \in \mathcal{S}_2(H) \quad \text{for some orthogonal } U. \end{aligned}$$

Theorem 1.1 replaces this orthogonal U by a coordinate symmetry $Q \in \mathcal{P}$. Covariance alone can no longer decide equivalence: an orthogonal mixing still has $TT^* = I$, but its non-Gaussian image is singular whenever it lies outside every such Hilbert–Schmidt neighbourhood. The new classification detects changes that the Gaussian covariance criterion cannot see.

Kakutani’s criterion encounters a different obstruction. Multiplying the one-coordinate affinities after mixing would compare μ with the product of the output marginals, rather than with their joint law ν_T . The energy $E_n(T)$ uses the full joint distribution of the first n outputs and retains their dependence. The theorem identifies boundedness of these energies with a Hilbert–Schmidt condition on T , without requiring the image to factor into independent coordinates or finite blocks.

Block rotations illustrate what Kakutani already settles. Rotating two independent Laplace variables gives an equivalent law because both densities are positive everywhere. Repeating a rotation through $\pi/4$ on disjoint pairs gives singularity by Kakutani’s theorem, applied to the pairs. For a general mixing there is no such independent block decomposition; Theorem 1.1 gives the operator criterion in its absence.

The theorem includes Laplace and e^{-x^4} densities after normalization, as well as asymmetric densities with an infinite fourth moment; examples appear at the end of Section 5. Two consequences distinguish the rigidity of exact symmetries from the regularity needed to control perturbations.

Corollary 1.2 (Moment-only stabilizer). *If ρ is non-Gaussian, centered, and has variance one, then the exact linear stabilizer of $\rho^{\otimes \mathbb{N}}$ is $\mathcal{P}$. No density or Fisher information assumption is needed.*

Corollary 1.3 (Orthogonal transformations). *If $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and f is non-Gaussian, centered, and has variance one, then every orthogonal U satisfies*

$$\nu_U \sim \mu \iff U - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}.$$

Rotations have no diagonal scale tangent, which is why location Fisher information suffices here.

1.2 Further context

Shimomura’s Theorem 3.4 shows that $I - S^2$ is compact for a self-adjoint admissible operator S , under ℓ^2 -continuity, ℓ^2 -translation quasi-invariance, and ℓ^2 -ergodicity; his examples preclude a universal Hilbert–Schmidt criterion under those assumptions alone [34]. The i.i.d. structure lets us go further: it identifies the coordinate symmetry approached by remote rows, and the local Hellinger geometry forces their errors to be square summable. This gives both necessity and sufficiency for arbitrary bounded invertible mixing. The classification complements constructions of non-Gaussian quasi-invariant measures [11, 38] by determining the linear transformations available to one fixed i.i.d. law.

The Fisher assumptions control the local cost of perturbing the density. For translations, Shepp relates this cost to regularity of the square-root density [32]; Euclidean and Lie-group perturbations lead to Fisher information as the local metric [24, 37]. Our proof uses this local metric to obtain a classification for arbitrary bounded invertible mixing. In the Gaussian setting, translation and nonlinear transformation laws are developed by Cameron–Martin and Ramer [7, 28]; here the issue is the rigidity imposed on linear mixing by a fixed non-Gaussian product.

Classical independence-of-linear-forms theorems [9, 15, 16, 36] characterize Gaussian laws through independence of suitable linear forms, while independent component analysis identifies non-Gaussian mixing matrices up to scaling and permutation [8]. These results explain why coordinate permutations appear among the exact symmetries. Measure equivalence allows dependent outputs, so our necessity proof must recover coordinate structure without assuming that the mixed variables are independent.

1.3 How tails and support change the answer

A large input can move many outputs at once. For stable coordinates this event is frequent enough to change the cost of mixing. The classification in [theorem 4.1](#) replaces the Hilbert–Schmidt condition by

$$\sum_i |m_{ii} - 1|^2 + \sum_j \left(\sum_{i \neq j} |m_{ij}|^2 \right)^{\alpha/2} < \infty.$$

Here $0 < \alpha < 2$ is the stable index and $M = TQ^{-1}$ is the matrix after removing a signed permutation; the theorem specifies the compatible row-series action. Scales retain a quadratic cost. An off-diagonal column records how far one input moves the other coordinates, and its ℓ^2 -length is charged to the power α . This distinguishes many independent small shears from a single source mixed into many outputs.

Zeros of the density create a different obstruction. A contraction of one coordinate can leave part of the original support with no preimage, even though its operator differs from the identity by rank one. [Theorem 5.2](#) separates this loss of support from the accumulated Hellinger cost: Hilbert–Schmidt proximity controls the latter, while the forward and inverse row sums must avoid the zeros of the density. For bounded or one-sided laws, the support alone forces the explicit permutation classifications of [theorem 5.3](#).

1.4 From a rotation to infinite-dimensional rigidity

The first question is how to detect mixing without demanding higher moments. Covariance cannot see a rotation, so we test it with bounded complex exponentials. Centering the two factors removes their marginal responses and leaves

$$R(\xi, \eta) = \eta \phi'(\xi) \phi(\eta) - \xi \phi(\xi) \phi'(\eta), \quad \phi(u) = \mathbb{E} e^{iuX}.$$

If every such response vanished, fixing one frequency would give a linear ODE forcing ϕ to be Gaussian. Geometrically, the weighted Fourier jet $u \mapsto (u\phi(u), \phi'(u))$ would lie in a line. A non-Gaussian law therefore supplies two independent Fourier directions. We use them to build tests dual to every matrix tangent.

An infinitesimal response is not yet a uniform lower bound. As the dimension grows, errors from many small entries could overwhelm it. We control these errors by following the square-root density along an exponential path and grouping terms according to their shared coordinates. Independence then turns the sums into Frobenius contractions. This gives a fixed local Hellinger bound in every dimension.

The remaining difficulty is global: why should an equivalent image law be close to a coordinate symmetry at all? Absolute continuity makes its remote marginals approach the original coordinate law. A characteristic-function argument shows that each corresponding row must approach an allowed signed coordinate. This identifies the axes, but says nothing yet about the sum of the errors.

To force summability, suppose that a fixed small amount of error could always be collected in the tail. Exchange those coordinates with later ones whose matching errors are negligible, and exchange the corresponding input coordinates as well. Compose the input exchange with the output exchange transported back through T . The resulting map differs from the identity by precisely the amount of matching error that the local lower bound detects. Yet absolute continuity makes every sufficiently remote exchange invisible in Hellinger distance. The contradiction

forces the matching errors to be square summable. Finite-coordinate approximation proves the converse, and translation ergodicity supplies the equivalence–singularity alternative.

We use A^* for the adjoint, P_n for coordinate projection on H , and $\|\cdot\|_F$ for the Frobenius norm.

2 Dimension-Free Hellinger Geometry

The Hilbert–Schmidt criterion suggests a local question: does the Hellinger distance created by a small matrix perturbation measure its Frobenius norm, with constants that survive as the dimension grows? The upper bound follows by integrating the Fisher length of a path. The lower bound requires a way to detect every direction of motion. We find it by first asking what a rotation does to two independent coordinates.

2.1 Detecting a rotation

Characteristic functions let us detect dependence with bounded tests. To remove the separate responses of the two marginals, we center each exponential before multiplying. For independent copies X_1, X_2 with $\mathbb{E}X = 0$ and $\mathbb{E}|X| < \infty$, write $\phi(u) = \mathbb{E}e^{iuX}$ and set

$$O_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad W_{\xi, \eta}(x, y) = (e^{i\xi x} - \phi(\xi))(e^{i\eta y} - \phi(\eta)).$$

Before rotation, independence gives

$$\mathbb{E}W_{\xi, \eta}(X_1, X_2) = 0, \quad \mathbb{E}|W_{\xi, \eta}(X_1, X_2)|^2 = (1 - |\phi(\xi)|^2)(1 - |\phi(\eta)|^2) \leq 1. \quad (6)$$

Also $|W_{\xi, \eta}| \leq 4$. Thus the test remains integrable after mixing, regardless of the tails. Its derivative at zero rotation answers the question suggested by Gaussian invariance: can a non-Gaussian law escape every one of these tests? The following infinitesimal form of Gaussian independence rigidity [15, 16] shows that it cannot.

Lemma 2.1 (Gaussian characterization by the rotational Fourier defect). *For the product of centered complex exponentials defined above,*

$$R(\xi, \eta) = \left. \frac{d}{d\theta} \mathbb{E}W_{\xi, \eta}(O_\theta(X_1, X_2)) \right|_{\theta=0} = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta). \quad (7)$$

Moreover,

$$R \equiv 0 \iff \text{Law}(X) \text{ is a centered Gaussian, possibly degenerate.} \quad (8)$$

Proof. Since $\mathbb{E}|X| < \infty$, we have $\phi \in C^1(\mathbb{R})$ and $\phi'(0) = i\mathbb{E}X = 0$. After rotation, the joint characteristic function is

$$\phi(\xi \cos \theta + \eta \sin \theta)\phi(-\xi \sin \theta + \eta \cos \theta).$$

The two marginal terms in the centered product have zero derivative at $\theta = 0$, again by $\phi'(0) = 0$; differentiating the displayed product gives (7).

To recover the law from a vanishing response, fix one frequency and regard the other as the variable of an ODE. Suppose $R \equiv 0$ and choose $\eta_0 \neq 0$ with $\phi(\eta_0) \neq 0$, possible because $\phi(0) = 1$. Then, for every ξ ,

$$\phi'(\xi) = c\xi\phi(\xi), \quad c = \frac{\phi'(\eta_0)}{\eta_0\phi(\eta_0)}, \quad \phi(0) = 1.$$

Uniqueness for this global linear ODE yields $\phi(\xi) = e^{c\xi^2/2}$. The identities $\phi(-\xi) = \overline{\phi(\xi)}$ and $|\phi(\xi)| \leq 1$ imply respectively $c \in \mathbb{R}$ and $c \leq 0$. Thus $c = -\sigma^2$ for some $\sigma \geq 0$, so X is centered Gaussian (the case $\sigma = 0$ is the point mass at zero). Conversely, direct substitution shows that every such Gaussian has $R \equiv 0$. $\square$

The response has a simple geometric interpretation. The two entries of the weighted Fourier jet

$$u \mapsto (u\phi(u), \phi'(u)) \in \mathbb{C}^2$$

form a curve whose two-point determinants are

$$\det \begin{pmatrix} \xi\phi(\xi) & \phi'(\xi) \\ \eta\phi(\eta) & \phi'(\eta) \end{pmatrix} = -R(\xi, \eta). \quad (9)$$

For a Gaussian this curve lies in a line; for every non-Gaussian it spans two directions. This is exactly the freedom we will need to separate the two ordered entries of a matrix tangent. The weight u comes from differentiating the exponential under rotation. It makes Gaussianity a linear dependence relation, and fixing one nonzero frequency proves that relation globally, even across zeros of ϕ .

Centering the response. Centering is what separates the effect of mixing from the motion of the mean. For a variable Y with mean m and characteristic function ϕ_Y , put $X = (Y - m)/\sigma$, where $\sigma > 0$. For $a = \xi/\sigma$ and $b = \eta/\sigma$, the rotational Fourier defect of X satisfies

$$R(\xi, \eta) = e^{-i(m/\sigma)(\xi+\eta)} [b\phi'_Y(a)\phi_Y(b) - a\phi_Y(a)\phi'_Y(b) + im(a-b)\phi_Y(a)\phi_Y(b)]. \quad (10)$$

The last term removes the contribution of the mean. Scaling then changes only the frequencies, apart from the phase introduced by centering.

2.2 The matrix Fisher operator

The rotation calculation identifies the exceptional law. To measure the size of a general infinitesimal mixing, we now compute its Fisher metric [1]. There are three natural motions: changing individual scales, mixing symmetrically, and rotating. The calculation below assigns one eigenvalue to each, and shows exactly where the Gaussian metric loses a direction. Write $g = \sqrt{f}$ and assume

$$f > 0 \text{ Lebesgue-almost everywhere,} \quad g \in W^{1,2}(\mathbb{R}), \quad xg'(x) \in L^2(\mathbb{R}), \quad (11)$$

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad X \sim f(x) dx.$$

The location score $s = -f'/f = -2g'/g$ satisfies

$$\mathbb{E}s(X) = 0, \quad \mathbb{E}[Xs(X)] = 1, \quad J := \mathbb{E}s(X)^2 = 4\|g'\|_2^2. \quad (12)$$

Define the scale score and its second moment by

$$d(x) := xs(x) - 1, \quad \kappa := \mathbb{E}d(X)^2 = 4\|xg'\|_2^2 - 1. \quad (13)$$

Here $f', xf' \in L^1$, so cutoff integration by parts against 1 and x gives the two identities in (12).

For independent copies $X_1, \dots, X_n$ of X , write $\mathbf{X}^{(n)} = (X_1, \dots, X_n) \sim \mu_n$. For $A \in M_n(\mathbb{R})$, define the score of the pushforward by $I + tA$ at $t = 0$ by

$$S_A(\mathbf{X}^{(n)}) = \sum_{i,j=1}^n A_{ij}(s(X_i)X_j - \delta_{ij}) = \sum_i A_{ii}d(X_i) + \sum_{i \neq j} A_{ij}s(X_i)X_j. \quad (14)$$

Theorem 2.2 (The matrix Fisher operator). *Under (11), let s and d be the location and scale scores defined above, and define the matrix Fisher operator $\mathcal{I}$ by $\langle A, \mathcal{I}B \rangle_F = \mathbb{E}[S_A S_B]$. Then*

$$\boxed{\mathcal{I}A = JA + A^* + (\kappa - J - 1) \text{diag } A.} \quad (15)$$

Here $\text{diag } A$ is the matrix with diagonal entries A_{ii} and zero off-diagonal entries. The diagonal, symmetric off-diagonal, and skew-symmetric matrix subspaces are orthogonal for the Frobenius inner product and have eigenvalues κ , $J + 1$, and $J - 1$, respectively. Equivalently,

$$\begin{aligned} \langle A, \mathcal{I}A \rangle_F &= \kappa \|\text{diag } A\|_F^2 + (J + 1) \left\| \frac{A + A^*}{2} - \text{diag } A \right\|_F^2 \\ &\quad + (J - 1) \left\| \frac{A - A^*}{2} \right\|_F^2. \end{aligned} \quad (16)$$

Their multiplicities are $n, n(n - 1)/2, n(n - 1)/2$, respectively. Moreover, $\kappa > 0$ and $J \geq 1$. Here $J = 1$ holds precisely for the standard Gaussian density. Thus, for a non-Gaussian f , $\min\{\kappa, J - 1\} > 0$ and

$$\begin{aligned} \min\{\kappa, J - 1\} \|A\|_F^2 &\leq \langle A, \mathcal{I}A \rangle_F \leq C_f \|A\|_F^2, \\ C_f &:= \max\{\kappa, J + 1\}. \end{aligned} \quad (17)$$

Proof. The variables $d(X_i)$ are centered and independent. Their covariances with the off-diagonal terms vanish, while two off-diagonal terms have nonzero covariance only when their ordered index pairs coincide or are reversed; the corresponding covariances are J and $(\mathbb{E}Xs(X))^2 = 1$. Hence

$$\mathbb{E}S_A^2 = \kappa \sum_i A_{ii}^2 + \sum_{i < j} \{J(A_{ij}^2 + A_{ji}^2) + 2A_{ij}A_{ji}\}. \quad (18)$$

Polarization gives (15); resolving each off-diagonal pair into its symmetric and skew-symmetric parts gives (16) and the stated multiplicities.

Cauchy–Schwarz applied to $\mathbb{E}Xs(X) = 1$ gives $J \geq 1$. Equality implies $s(X) = X$ almost surely, hence $g' = -xg/2$ almost everywhere and f is the standard Gaussian. If $\kappa = 0$, then $2xg' + g = 0$ almost everywhere on each of the two half-lines. The absolutely continuous solutions there are constant multiples of $|x|^{-1/2}$, which cannot give a positive probability density. This proves $\kappa > 0$ and the asserted bounds. $\square$

2.3 Following square-root densities

The rotational eigenvalue $J - 1$ is the part of the Fisher metric that covariance misses. Together with the scale and symmetric eigenvalues, it measures every matrix tangent when the law is non-Gaussian. To turn this infinitesimal calculation into a comparison of measures, we work with square-root densities: their ordinary L^2 distance is Hellinger distance, and linear changes of variables act unitarily. This is the quadratic-mean viewpoint of [39, Section 7.2]; see also [3, 5].

For $A \in \text{GL}(n, \mathbb{R})$ define a unitary operator on $L^2(\mathbb{R}^n, dx)$ by

$$(U_A v)(y) = |\det A|^{-1/2} v(A^{-1}y), \quad \Omega_n(x) = \prod_{i=1}^n g(x_i).$$

Thus $U_A \Omega_n$ is the square root of the Lebesgue density of $A_{\#} \mu_n$. For invertible A, B ,

$$h(A_{\#} \mu_n, B_{\#} \mu_n) = \|U_A \Omega_n - U_B \Omega_n\|_2.$$

For a fixed real matrix A , the generator of $U_{\exp(tA)}$ is

$$G_A v = -\frac{1}{2} \text{tr}(A) v - (Ax) \cdot \nabla v, \quad G_A \Omega_n = \frac{1}{2} S_A \Omega_n. \quad (19)$$

These identities hold in L^2 : the diagonal terms use $xg' \in L^2$, the off-diagonal terms use $g' \in L^2$ and $\mathbb{E}X^2 = 1$, and cutoff followed by mollification gives convergence in the required weighted Sobolev graph norm.

Lemma 2.3 (A dimension-free Hellinger path bound). *If A_t , $0 \leq t \leq 1$, is a continuously differentiable path in $\text{GL}(n, \mathbb{R})$, then $t \mapsto U_{A_t} \Omega_n$ is continuously differentiable in L^2 and*

$$\begin{aligned} \frac{d}{dt} U_{A_t} \Omega_n &= U_{A_t} G_{A_t^{-1} \dot{A}_t} \Omega_n, \\ \left\| \frac{d}{dt} U_{A_t} \Omega_n \right\|_2 &= \frac{1}{2} \sqrt{\left\langle A_t^{-1} \dot{A}_t, \mathcal{I}(A_t^{-1} \dot{A}_t) \right\rangle_F}. \end{aligned} \quad (20)$$

Consequently,

$$h((A_1)_{\#} \mu_n, (A_0)_{\#} \mu_n) \leq \frac{\sqrt{C_f}}{2} \int_0^1 \|A_t^{-1} \dot{A}_t\|_F dt. \quad (21)$$

In particular, if $\|B\|_{\text{op}} \leq r < 1$, then

$$h((I + B)_{\#} \mu_n, \mu_n) \leq \frac{\sqrt{C_f}}{2(1-r)} \|B\|_F. \quad (22)$$

All constants are independent of n .

Proof. The identity $U_A U_B = U_{AB}$ and (19) give the derivative formula. Unitarity and (17) give its norm; integration proves (21). For the last bound take $A_t = I + tB$ and use $\|A_t^{-1}\|_{\text{op}} \leq (1-r)^{-1}$. $\square$

2.4 Bounded Fourier tests for matrix tangents

We now return to a non-Gaussian f and seek a lower bound. Integrating the speed of a path gives an upper bound on the distance travelled; a lower bound must also prevent cancellation along the path. We do this by pairing the displacement with a test chosen for its direction.

The score S_A suggests the right pairing, but following the score itself would introduce derivatives and moments beyond our assumptions. The Fourier jet gives a smoother choice. Each off-diagonal score term is $s(X_i)X_j$. We choose bounded Fourier sums p, q dual to s, x , so that $p(X_i)q(X_j)$ detects the ordered entry A_{ij} and cancels the reversed entry. A third Fourier sum detects changes of scale.

Lemma 2.4 (Finite real Fourier sums with dual moments). *There exist centered finite real Fourier sums p, q, r such that*

$$\begin{array}{c|cc} & s & x \\ p & 1 & 0 \\ q & 0 & 1 \end{array} \quad \text{and} \quad \mathbb{E}[r(X)d(X)] = 1. \quad (23)$$

The entries in the table denote expectations of the products. For

$$H_A(\mathbf{X}^{(n)}) := \sum_i A_{ii}r(X_i) + \sum_{i \neq j} A_{ij}p(X_i)q(X_j), \quad (24)$$

one has, for all real matrices A, B of the same size,

$$\mathbb{E}[H_A S_B] = \langle A, B \rangle_F, \quad \|H_A\|_{L^2(\mu_n)} \leq C_{\text{dual}} \|A\|_F, \quad (25)$$

where C_{dual} is independent of n .

Proof. On the real space of centered finite real Fourier sums, consider

$$L_1(u) = \mathbb{E}u'(X) = \mathbb{E}[u(X)s(X)], \quad L_2(u) = \mathbb{E}[Xu(X)].$$

The integration-by-parts identity is valid because u, u' are bounded. If $u \mapsto (L_1(u), L_2(u))$ had rank at most one, then after complexification its determinant on the centered exponentials of frequencies ξ, η would vanish. Since $L_1(e^{i\xi x}) = i\xi\phi(\xi)$ and $L_2(e^{i\xi x}) = -i\phi'(\xi)$, this would give

$$\eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta) = 0 \quad (\xi, \eta \in \mathbb{R}).$$

By lemma 2.1, this contradicts non-Gaussianity. Thus the moment map has rank two, and inverting the moment matrix of two Fourier sums gives p, q with the required dual moments.

For $\xi \neq 0$ let $r_0(x) = \cos(\xi x) - \mathbb{E}\cos(\xi X)$. Dominated convergence gives

$$\frac{\mathbb{E}[Xr_0'(X)]}{\xi^2} = -\mathbb{E}\left[X\frac{\sin(\xi X)}{\xi}\right] \longrightarrow -\mathbb{E}X^2 = -1 \quad (\xi \rightarrow 0).$$

For small nonzero ξ the denominator is nonzero, so $r = r_0/\mathbb{E}[Xr_0'(X)]$ satisfies, by cutoff integration by parts, $\mathbb{E}[r(X)d(X)] = \mathbb{E}[Xr'(X)] = 1$.

Centering kills all diagonal–off-diagonal cross terms. Coincident ordered pairs contribute $\mathbb{E}[ps]\mathbb{E}[qX] = 1$, whereas reversed pairs contribute $\mathbb{E}[pX]\mathbb{E}[qs] = 0$; hence $\mathbb{E}[H_A S_B] = \langle A, B \rangle_F$. Finally, independence gives, for centered $a, b \in L^2(f)$,

$$\left\| \sum_{i \neq j} A_{ij}a(X_i)b(X_j) \right\|_2 \leq \sqrt{2} \|a\|_2 \|b\|_2 \|A\|_F. \quad (26)$$

Combining this with orthogonality of the centered diagonal terms proves the uniform norm bound. $\square$

The dual pairing is exact at the identity. To use it along a path, we must control how fast the test itself moves. The apparent obstacle is the number of terms in $H_A S_B$, which grows like the fourth power of the dimension. Most cross terms vanish by centering. Grouping the others by their shared coordinates leaves a fixed number of patterns, each controlled by a Frobenius contraction. This is where independence makes the estimate uniform in the dimension.

Lemma 2.5 (Generator bound for the functions H_A). *Let p, q, r be any fixed centered bounded continuously differentiable functions with bounded derivatives. Define H_A by (24). There is a constant C_{gen} , depending on these functions and on J, κ , such that*

$$\|G_B(H_A \Omega_n)\|_{L^2(dx)} \leq C_{\text{gen}} \|A\|_F \|B\|_F \quad (A, B \in M_n(\mathbb{R}), n \geq 1). \quad (27)$$

Proof. For fixed n , boundedness of H_A and its first derivatives, together with $\mathbb{E}X^2 = 1$ and the weighted Sobolev regularity of Ω_n , puts $H_A \Omega_n$ in the domain of G_B . The product rule gives

$$G_B(H_A \Omega_n) = \left\{ \frac{1}{2} H_A S_B - (Bx) \cdot \nabla H_A \right\} \Omega_n. \quad (28)$$

If $u_1, \dots, u_m$ are centered $L^2(f)$ functions and C is indexed by ordered tuples of *distinct* coordinates, then

$$\left\| \sum_{i_1, \dots, i_m}^{\neq} C_{i_1 \dots i_m} \prod_{a=1}^m u_a(X_{i_a}) \right\|_2 \leq \sqrt{m!} \|C\|_{\ell^2} \prod_{a=1}^m \|u_a\|_2. \quad (29)$$

After squaring, only pairs of tuples with the same index set survive; summing their at most $m!$ matchings proves the estimate. We use it for $m \leq 4$.

For a vector a and a zero-diagonal matrix M , abbreviate

$$D_a(u) = \sum_i a_i u(X_i), \quad Q_M(u, v) = \sum_{i \neq j} M_{ij} u(X_i) v(X_j).$$

Then $H_A S_B$ is obtained by multiplying $D_{\mathbf{a}}(r) + Q_{A_0}(p, q)$ and $D_{\mathbf{b}}(d) + Q_{B_0}(s, x)$, where $\mathbf{a} = (A_{ii})_i$, $\mathbf{b} = (B_{ii})_i$, $A_0 = A - \text{diag } A$, and $B_0 = B - \text{diag } B$. Every local product on a shared coordinate is a bounded function times one of d, s, x , hence is in $L^2(f)$; split it into its mean and centered part. For $D_a D_b$, the distinct-index tensor, Hadamard vector, and scalar contraction have norm at most $\|a\|_2 \|b\|_2$. For $D_a Q_M$, the three coincidence classes yield $a \otimes M$, a row or column multiplication of M , and Ma or $M^* a$, all bounded by $\|a\|_2 \|M\|_F$.

For the remaining product

$$Q_M(u, v) Q_N(w, z) = \sum_{i \neq j, k \neq l} M_{ij} N_{kl} u_i v_j w_k z_l,$$

with u, v bounded and $w, z \in L^2$, four distinct indices give $M \otimes N$. In the representative one-coincidence pattern $i = k$, the centered coefficient has norm

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_l N_{il}^2 \right) \right\}^{1/2} \leq \|M\|_F \|N\|_F.$$

Its mean gives the off-diagonal part of M^*N ; the other one-coincidence classes are transposes. When both indices are shared, Cauchy–Schwarz bounds the Hadamard coefficients $M_{ij}N_{ij}$, $M_{ij}N_{ji}$, and all their row, column, and scalar contractions by $\|M\|_F\|N\|_F$. These are all partitions of the two ordered pairs. Applying (29) to the centered pieces proves

$$\|H_A S_B\|_2 \leq C\|A\|_F\|B\|_F. \quad (30)$$

For $(Bx) \cdot \nabla H_A$, the derivative of the diagonal part is

$$\sum_{i,k} A_{ii} B_{ik} r'(X_i) X_k.$$

After centering r' and xr' , its coefficient arrays are $(\text{diag } A)B$, $B^* \mathbf{a} - (a_i b_i)_i$, and a diagonal contraction; their norms are at most $2\|A\|_F\|B\|_F$.

With $M = A_0$, the off-diagonal derivative is the sum of

$$F(M, B; p', q) := \sum_{i \neq j, k} M_{ij} B_{ik} p'(X_i) q(X_j) X_k$$

and $F(M^*, B; q', p)$. For $k \notin \{i, j\}$, centering p' gives the three-index bound

$$\left\{ \sum_i \left(\sum_j M_{ij}^2 \right) \left(\sum_k B_{ik}^2 \right) \right\}^{1/2} \leq \|M\|_F \|B\|_F,$$

and its mean gives $M^*B - M^* \text{diag } B$. The classes $k = i$ and $k = j$ give $(B_{ii}M_{ij})$, $M^* \mathbf{b}$, $(M_{ij}B_{ij})_{ij}$, and their row, column, and scalar contractions. The same Frobenius estimates control each array, while boundedness of p', q and $\mathbb{E}X^2 = 1$ puts every local factor in L^2 . Applying the same argument to $F(M^*, B; q', p)$ and then (29) yields

$$\|(Bx) \cdot \nabla H_A\|_2 \leq C\|A\|_F\|B\|_F.$$

Together with (30) and (28), this proves the lemma. $\square$

We can now keep the first-order response from cancelling. Pairing with H_A gives a quadratic main term, while the generator bound makes the error cubic. Choosing one sufficiently small radius therefore works in every dimension.

Proposition 2.6 (Uniform local Hellinger coercivity). *For every non-Gaussian density satisfying (11), there are constants $c, C, \delta > 0$, independent of n , such that*

$$c\|B\|_F \leq h((I + B)_{\#} \mu_n, \mu_n) \leq C\|B\|_F \quad \text{whenever } \|B\|_F \leq \delta. \quad (31)$$

The lower bound can be obtained using the fixed finite set of coordinate frequencies from lemma 2.4.

Proof. Fix the finite real Fourier sums from lemma 2.4. In exponential coordinates, unitarity and the group derivative give

$$\langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle = \int_0^1 \langle U_{\exp(-tA)}(H_A \Omega_n), G_A \Omega_n \rangle dt.$$

The value obtained by replacing $U_{\exp(-tA)}(H_A\Omega_n)$ with $H_A\Omega_n$ is $\frac{1}{2}\mathbb{E}H_AS_A = \frac{1}{2}\|A\|_F^2$. Moreover, $\|(U_{\exp(-tA)} - I)v\|_2 \leq t\|G_A v\|_2$ on the generator domain. Therefore [lemma 2.5](#) and (17) yield

$$\left| \langle H_A\Omega_n, (U_{\exp A} - I)\Omega_n \rangle - \frac{1}{2}\|A\|_F^2 \right| \leq \frac{C_{\text{gen}}\sqrt{C_f}}{4}\|A\|_F^3.$$

For sufficiently small $\|A\|_F$, Cauchy–Schwarz and (25) thus give

$$h((\exp A)_{\#}\mu_n, \mu_n) \geq \frac{1}{4C_{\text{dual}}}\|A\|_F.$$

For $\|B\|_{\text{op}} < 1/2$, $A = \log(I + B)$ is real and $\|A - B\|_F \leq C\|B\|_{\text{op}}\|B\|_F$. Hence $\|A\|_F$ and $\|B\|_F$ are uniformly comparable in a fixed chart, proving the lower bound. The upper bound is [lemma 2.3](#). $\square$

2.5 The global comparison

Near the identity, Hellinger distance now measures Frobenius distance. The same is true near every allowed coordinate symmetry, since these preserve the law. The global question is whether matrices far from all such symmetries could nevertheless change the law by arbitrarily little. The infinite-dimensional classification will rule this out: stacking such matrices in independent blocks would contradict Hilbert–Schmidt necessity. We state the resulting comparison here and prove it in [Section 3.6](#).

Theorem 2.7 (Dimension-free global Hellinger coercivity). *Assume (1). Let $\mathcal{P}_n$ denote the allowed signed-permutation matrices in dimension n , and set*

$$d_n(T) = \min_{Q \in \mathcal{P}_n} \|T - Q\|_F. \quad (32)$$

For every $0 < a \leq b < \infty$ there are constants $c = c(f, a, b) > 0$ and $C = C(f, a, b) < \infty$ such that, for every n and every $T \in \text{GL}(n, \mathbb{R})$ whose singular values lie in $[a, b]$,

$$c \min\{d_n(T)^2, 1\} \leq h(\mu_n, T_{\#}\mu_n)^2 \leq C \min\{d_n(T)^2, 1\}. \quad (33)$$

3 The Finite-Variance Classification

The local estimate answers how much a small perturbation changes the law. The classification asks the converse question: how can equivalence of laws force a general operator to be a small perturbation of a coordinate symmetry in the tail? Absolute continuity will identify the individual axes. The decisive step is then to compare errors on one group of coordinates with negligible errors on another, using exchanges that preserve the product law.

3.1 Making row sums composable

We will compare a mixing with its inverse and with conjugated coordinate exchanges. These operations require actual maps on the product space. A sample almost surely lies outside ℓ^2 , even though every row sum is well defined in L^2 . The way to make the maps composable is to choose the subsequence defining each sum from its coefficients alone. The same choice then works before and after a bounded linear mixing. This realization of Hilbert-space operators on the product space goes back to [\[34\]](#). Throughout this construction, $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$.

Lemma 3.1 (Measurable row-series realization on each countable operator orbit). *For every $T \in \mathcal{B}(H)$, there is a measurable row-series realization, namely a Borel map $\Phi_T : \mathbb{R}^{\mathbb{N}} \rightarrow \mathbb{R}^{\mathbb{N}}$ whose coordinates are the $L^2(\mu)$ row sums*

$$(\Phi_T x)_i = \sum_{j \geq 1} T_{ij} x_j.$$

These maps can be chosen so that the following statements hold. If a probability measure λ satisfies, for some finite $C_\lambda \geq 0$,

$$\int \left| \sum_j a_j x_j \right|^2 d\lambda(x) \leq C_\lambda \|a\|_2^2 \quad \text{for every finitely supported } a, \quad (34)$$

then $(\Phi_T)_\# \lambda$ also satisfies this condition. Its constant is at most $C_\lambda \|T\|_{\text{op}}^2$. Also

$$\Phi_S \circ \Phi_T = \Phi_{ST} \quad \lambda\text{-almost surely} \quad (S, T \in \mathcal{B}(H)). \quad (35)$$

For each fixed countable subgroup $\Gamma \leq \text{GL}(H)$, there is a Γ -invariant Borel set $D \subset \mathbb{R}^{\mathbb{N}}$ of full measure for every λ satisfying (34). On this set, these restrictions form an action by bimeasurable automorphisms. In particular, this applies to μ and all its Γ -orbit laws, including singular orbit laws.

Proof. For $a \in \ell^2$, the finite sums $\sum_{j \leq n} a_j x_j$ have a unique $L^2(\lambda)$ limit L_a . Choose

$$n_k(a) = \min \left\{ n \geq k : \sum_{j > n} |a_j|^2 \leq 2^{-4k} \right\}.$$

Define $\hat{L}_a(x)$ to be the limit of $\sum_{j \leq n_k(a)} a_j x_j$ when it exists, and zero otherwise. By (34),

$$\left\| \sum_{j \leq n_k(a)} a_j x_j - L_a \right\|_{L^2(\lambda)}^2 \leq C_\lambda 2^{-4k}.$$

Chebyshev's inequality and Borel–Cantelli give convergence along this subsequence λ -almost surely and in $L^2(\lambda)$. This deterministic choice works for every law satisfying (34). Set $\Phi_T(x)_i = \hat{L}_{T^* e_i}(x)$; the construction is Borel.

For finitely supported b , $\sum_i b_i \Phi_T(x)_i = L_{T^* b}$ almost surely, proving the asserted second-moment bound for the pushforward. For the i -th row of S ,

$$\sum_{j \leq n} S_{ij} \hat{L}_{T^* e_j} = L_{T^* P_n S^* e_i} \longrightarrow L_{T^* S^* e_i} \quad \text{in } L^2(\lambda).$$

The selected subsequence on the left also converges almost surely to $\Phi_S(\Phi_T(x))_i$, because the output law satisfies the same type of bound. Uniqueness of the limit proves (35).

For a countable Γ , let B be the Borel set on which all its row limits, identity relations, and pairwise composition relations hold. The preceding argument gives $\lambda(B) = 1$ for every admissible law. Put

$$D = \bigcap_{T \in \Gamma} \Phi_T^{-1}(B).$$

Every orbit law satisfies (34), so D has full measure. The composition identities make D invariant and $\Phi_{T^{-1}}$ the inverse of Φ_T on D . $\square$

Write $\nu_T = (\Phi_T)_\# \mu$.

3.2 Why overlap implies equivalence

Two arbitrary measures can overlap without being equivalent. For the positive-density products considered here, translations rule out this intermediate possibility. Both the original and the mixed law admit ℓ^2 shifts, and neither can be split into nontrivial pieces invariant under those shifts. Their common part must therefore be all or nothing. This is the mechanism of Okazaki's dichotomy [27]. The product translation theory is developed in [2, 13, 14, 29, 32]; we give the argument needed here.

Lemma 3.2 (Equivalence–singularity dichotomy on every bounded invertible orbit). *For every $T \in \text{GL}(H)$,*

$$\nu_T \sim \mu \quad \text{or} \quad \nu_T \perp \mu. \quad (36)$$

The assumptions are $f > 0$ a.e., $\sqrt{f} \in W^{1,2}(\mathbb{R})$, $\mathbb{E}X = 0$, and $\mathbb{E}X^2 = 1$.

Proof. Put $g = \sqrt{f}$. The translation estimate

$$1 - A(f, f(\cdot - a)) = \frac{1}{2} \|g(\cdot - a) - g\|_2^2 \leq \frac{1}{2} \|g'\|_2^2 a^2$$

and Kakutani's theorem imply $(x \mapsto x + a)_\# \mu \sim \mu$ for every $a \in \ell^2$. Since the scalar pairing of a row of T with $T^{-1}a$ is absolutely convergent,

$$\Phi_T(x + T^{-1}a) = \Phi_T(x) + a \quad \mu\text{-almost surely};$$

hence ν_T is also quasi-invariant under every ℓ^2 translation.

Let $\mathcal{A}$ be the countable additive group generated by $\{qe_j, qTe_j : q \in \mathbb{Q}, j \geq 1\}$. Both μ and ν_T are quasi-invariant and ergodic under translation by $\mathcal{A}$. Indeed, rational-translation invariance and positivity of the finite-dimensional densities make every finite-coordinate section constant almost everywhere (convolve with a smooth approximate identity); the event is therefore a tail event, so Kolmogorov's zero–one law applies. Pullback by Φ_T proves the same assertion for ν_T , because translation by qTe_j pulls back to translation by qe_j .

Write $\nu_T = (\nu_T)_{ac} + (\nu_T)_s$ for its Lebesgue decomposition relative to μ . Choose a Borel μ -null set N_0 supporting $(\nu_T)_s$, and let $N = \bigcup_{a \in \mathcal{A}} (N_0 + a)$. Quasi-invariance of μ makes N a μ -null set; it is exactly $\mathcal{A}$ -invariant and still supports $(\nu_T)_s$. Thus $\nu_T(N) = (\nu_T)_s(\mathbb{R}^{\mathbb{N}}) \in \{0, 1\}$ by ergodicity. Mass one gives singularity. If the mass is zero, write $\nu_T = \ell_T \mu$. Quasi-invariance makes $\{\ell_T > 0\}$ invariant modulo μ ; it has positive measure and hence, by ergodicity, full measure. Thus $\nu_T \sim \mu$. $\square$

3.3 Reading the full law through finite marginals

Every finite output marginal has a positive density. Singularity can therefore emerge only as more coordinates are observed. The likelihood ratios form a martingale, and the limiting affinity measures the overlap that survives these observations. By the preceding dichotomy, any positive overlap gives equivalence. The likelihood interpretation is standard [19, 22]; here it lets us compare dependent outputs, whereas independent additive perturbations can be studied through convolution [18].

Lemma 3.3 (Projected Hellinger energy and finite-marginal likelihood ratios). *Assume $f > 0$ almost everywhere, $\sqrt{f} \in W^{1,2}(\mathbb{R})$, and $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. For $T \in \text{GL}(H)$, every finite output marginal $(\pi_n)_\# \nu_T$ is equivalent to μ_n . Writing $\mathcal{F}_n = \sigma(x_1, \dots, x_n)$, define*

$$\ell_{T,n}(x) = \frac{d[(\pi_n)_\# \nu_T]}{d\mu_n}(\pi_n x). \quad (37)$$

Then $A(\mu_n, (\pi_n)_\# \nu_T) = \mathbb{E}_\mu \sqrt{\ell_{T,n}}$, and $E_n(T)$ is as in (4); moreover, $(\ell_{T,n})_{n \geq 1}$ is a nonnegative mean-one μ -martingale. We have

$$A(\mu_n, (\pi_n)_\# \nu_T) \downarrow A(\mu, \nu_T), \quad \nu_T \sim \mu \iff \sup_n E_n(T) < \infty. \quad (38)$$

Otherwise $E_n(T) \uparrow \infty$ and $\nu_T \perp \mu$. If $\nu_T \sim \mu$ with density $\ell_T = d\nu_T/d\mu$, then

$$\ell_{T,n} = \mathbb{E}_\mu[\ell_T | \mathcal{F}_n] \longrightarrow \ell_T \quad \text{in } L^1(\mu), \quad (39)$$

$$\ell_{T,n}^{-1} = \mathbb{E}_{\nu_T}[\ell_T^{-1} | \mathcal{F}_n] \longrightarrow \ell_T^{-1} \quad \text{in } L^1(\nu_T). \quad (40)$$

In particular, both likelihood-ratio martingales are uniformly integrable in their respective reference measures.

Proof. The first n rows of T are linearly independent, since T^* is injective, so some n input columns form an invertible minor A . Writing the first n outputs as $AU + W$, with $U \sim \mu_n$ independent of W , shows that their density is a mixture of translates of the a.e.-positive density of AU . It is positive and finite almost everywhere, proving finite-dimensional equivalence.

Consistency of projected laws proves the martingale assertion. The martingale limit is $\ell_{T,\infty} = d(\nu_T)_{ac}/d\mu$, the density of the absolutely continuous part of ν_T . Indeed, Fatou on cylinder sets gives $\ell_{T,\infty}\mu \leq \nu_T$; conversely, if $(\nu_T)_{ac} = \ell_{ac}\mu$, then $\mathbb{E}_\mu[\ell_{ac} | \mathcal{F}_n] \leq \ell_{T,n}$, and martingale convergence gives the reverse inequality. Since $\mathbb{E}_\mu(\sqrt{\ell_{T,n}})^2 = 1$, the square roots are uniformly integrable and converge in L^1 to $\sqrt{\ell_{T,\infty}}$. Conditional Jensen also makes their expectations decreasing; hence

$$\lim_n A(\mu_n, (\pi_n)_\# \nu_T) = \int \sqrt{\ell_{T,\infty}} d\mu = A(\mu, \nu_T).$$

The dichotomy in lemma 3.2 now proves (38) and the singular alternative.

Under equivalence, the forward identity is the defining conditional-expectation identity for a marginal density. For the reverse, for $B \in \mathcal{F}_n$,

$$\int_B \ell_{T,n}^{-1} d\nu_T = \mu(B) = \int_B \ell_T^{-1} d\nu_T.$$

Conditional-expectation convergence gives both L^1 limits and hence both uniform-integrability assertions. $\square$

Since initial segments are cofinal among finite coordinate sets, Hellinger data processing also makes the criterion independent of their enumeration [21].

3.4 Sufficiency of Hilbert–Schmidt perturbations

For a Hilbert–Schmidt perturbation, the finite-dimensional path bound is already enough. Truncate the matrix to finitely many coordinates. The remaining Hilbert–Schmidt norm tends to zero, so the square-root likelihoods are Cauchy and converge to a density for the mixed law. Applying the same argument to the inverse gives the reverse absolute continuity.

Proposition 3.4 (Hilbert–Schmidt sufficiency and likelihood-ratio convergence). *Assume (11) and let $T = I + K \in \text{GL}(H)$ with $K \in \mathcal{S}_2$. Then $\nu_T \sim \mu$. For the eventually invertible operators $T_n = I + P_n K P_n$, their likelihood ratios $\ell_{T_n} = d\nu_{T_n}/d\mu$ converge in $L^1(\mu)$ to the strictly positive likelihood ratio $\ell_T = d\nu_T/d\mu$, and $\sqrt{\ell_{T_n}} \rightarrow \sqrt{\ell_T}$ in $L^2(\mu)$. The inverse approximants likewise satisfy $\ell_{T_n^{-1}} \rightarrow \ell_{T^{-1}}$ in $L^1(\mu)$. If Q is an exact linear stabilizer of μ , then $\nu_T \sim \mu$ whenever $TQ^{-1} - I \in \mathcal{S}_2$ and T is invertible.*

Proof. Since $P_n K P_n \rightarrow K$ in Hilbert–Schmidt norm, $T_n \rightarrow T$ in operator norm, so the approximants are eventually invertible with uniformly bounded inverse norms; their finite-coordinate likelihood ratios ℓ_{T_n} are positive almost everywhere. For all sufficiently large m, n , the operator $T_n^{-1} T_m - I$ acts in a finite coordinate block and has operator norm at most $1/2$. Hellinger invariance under the finite invertible map T_n^{-1} and (22) give

$$\|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2 \leq \sqrt{C_f} \|T_n^{-1}\|_{\text{op}} \|T_m - T_n\|_{\mathcal{S}_2}. \quad (41)$$

Thus $\sqrt{\ell_{T_n}}$ converges in $L^2(\mu)$ to a nonnegative function of norm one; denote its square by $\tilde{\ell}_T$. Since

$$\|\ell_{T_m} - \ell_{T_n}\|_1 \leq 2\|\sqrt{\ell_{T_m}} - \sqrt{\ell_{T_n}}\|_2,$$

we also have $\ell_{T_n} \rightarrow \tilde{\ell}_T$ in L^1 . Meanwhile, $\Phi_{T_n} \mathbf{X} \rightarrow \Phi_T \mathbf{X}$ in L^2 on every finite set of output coordinates. Cylinder tests therefore identify $\ell_T \mu$ with ν_T .

The inverse approximants converge in Hilbert–Schmidt norm because

$$\|T_n^{-1} - T^{-1}\|_{\mathcal{S}_2} \leq \|T_n^{-1}\|_{\text{op}} \|T_n - T\|_{\mathcal{S}_2} \|T^{-1}\|_{\text{op}}.$$

Applying the same argument gives $\nu_{T^{-1}} \ll \mu$. On the common measurable carrier supplied by lemma 3.1,

$$\mu = (\Phi_T)_{\#} \nu_{T^{-1}} \ll (\Phi_T)_{\#} \mu = \nu_T.$$

This proves equivalence and identifies $\tilde{\ell}_T = \ell_T > 0$ almost everywhere. Finally, $\nu_Q = \mu$ and $T = (TQ^{-1})Q$ give the assertion about exact stabilizers. $\square$

3.4.1 Rotations and the scale score

An orthogonal path has a skew-symmetric tangent, whose diagonal vanishes. The scale score therefore disappears from both the path length and the Fourier pairing. The assumptions reduce to

$$f > 0 \text{ almost everywhere, } g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1. \quad (42)$$

Cutoff integration by parts gives $\mathbb{E}s(X) = 0$, $\mathbb{E}[Xs(X)] = 1$, and, for skew-symmetric A ,

$$\mathbb{E}S_A^2 = (J - 1)\|A\|_F^2. \quad (43)$$

Only $x_j \partial_i \Omega_n$ with $i \neq j$ occur in the generator formula, which follows by radial cutoff and mollification commuting with the orthogonal one-parameter group.

Proposition 3.5 (Orthogonal local bounds and Hilbert–Schmidt sufficiency). *Under (42), every orthogonal U on ℓ^2 with $U - I \in \mathcal{S}_2$ satisfies $\nu_U \sim \mu$. Its likelihood ratios have the same L^1 and square-root L^2 convergence along finite-coordinate orthogonal approximants, as do the inverse likelihood ratios. If f is non-Gaussian, there also exist dimension-free $c, C, \delta > 0$ such that*

$$c\|V - I\|_F \leq h(V_{\#} \mu_n, \mu_n) \leq C\|V - I\|_F \quad (V \in O(n), \|V - I\|_F \leq \delta). \quad (44)$$

Proof. For a skew-symmetric A , integration along $\exp(tA)$ gives

$$h((\exp A)_{\#} \mu_n, \mu_n) \leq \frac{1}{2} \sqrt{J - 1} \|A\|_F. \quad (45)$$

The construction of p, q in [lemma 2.4](#) uses only $s(X), X \in L^2$ and $J-1 > 0$. For skew-symmetric A , take $H_A = \sum_{i \neq j} A_{ij} p(X_i) q(X_j)$. Both H_A and S_A have no diagonal term, so the generator proof uses only bounded functions times $s(X)$ or X and remains valid under [\(42\)](#). The half-density pairing gives the local lower bound in exponential coordinates. Since a sufficiently small orthogonal $V - I$ has a real skew-symmetric logarithm with $\|\log V\|_F \asymp \|V - I\|_F$, this proves [\(44\)](#).

Truncation does not preserve orthogonality. We restore it by taking the polar factor of $T_n = I + P_n(U - I)P_n$:

$$V_n = T_n(T_n^* T_n)^{-1/2}.$$

It is orthogonal and acts identically outside the first n coordinates. Since $T_n \rightarrow U$ in Hilbert–Schmidt norm and its least singular value is eventually bounded below by some $a > 0$,

$$\|V_n - T_n\|_{\mathcal{S}_2} = \|I - |T_n|\|_{\mathcal{S}_2} \leq \frac{\|T_n^* T_n - I\|_{\mathcal{S}_2}}{1 + a}.$$

The right side tends to zero, so $V_n \rightarrow U$ in Hilbert–Schmidt norm. For large m, n , apply [\(45\)](#) to the small skew-symmetric logarithm of $V_n^* V_m$. The square roots of the likelihood ratios are Cauchy, and cylinder tests identify their limit as in [proposition 3.4](#). Repeating the argument for $V_n^{-1} \rightarrow U^{-1}$ and using the measurable inverse action proves equivalence and all asserted convergence. $\square$

3.5 Necessity of Hilbert–Schmidt perturbations

We now start from equivalence of laws, with no prescribed coordinate symmetry. Before locating that symmetry, we pass the local comparison to Hilbert–Schmidt operators. This will let us apply the finite-dimensional estimate to the conjugated exchanges at the end of the argument.

Lemma 3.6 (The local bound for Hilbert–Schmidt operators). *Let f be non-Gaussian and satisfy [\(11\)](#). There are constants $c, C, \delta > 0$ such that every $K \in \mathcal{S}_2(\ell^2)$ with $\|K\|_{\mathcal{S}_2} \leq \delta$ satisfies*

$$c\|K\|_{\mathcal{S}_2} \leq h(\nu_{I+K}, \mu) \leq C\|K\|_{\mathcal{S}_2}. \quad (46)$$

The constants can be chosen with $\delta < 1$, so $I + K$ is automatically invertible. Under the weaker assumptions [\(42\)](#), the same conclusion holds for orthogonal $I + K$ whenever f is non-Gaussian.

Proof. Let $K_n = P_n K P_n$. Since the untouched tail has affinity one, [\(31\)](#) applies to $I + K_n$, while [proposition 3.4](#) gives

$$h(\nu_{I+K_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|K_n\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2}.$$

Passing to the limit proves both bounds.

For the orthogonal assertion, choose the radius in this lemma to be half the finite-dimensional radius in [\(44\)](#). The polar approximants from [proposition 3.5](#) satisfy

$$h(\nu_{V_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|V_n - I\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2},$$

Thus all sufficiently large approximants lie within the finite-dimensional radius, and [\(44\)](#) passes to the limit. $\square$

3.5.1 Finding the coordinate axes

Why should a row whose output resembles X concentrate on a single coordinate? If its coefficients are all small, their sum is approximately Gaussian. More generally, large coefficients survive in a subsequential limit and the remaining small coefficients contribute a Gaussian summand. A non-Gaussian law can reproduce itself in this way only when one surviving coefficient is an allowed sign. The next two lemmas prove this using just $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and non-Gaussianity.

Lemma 3.7 (A product approximation for small coefficients). *There is a nonnegative function ω_ρ , with $\omega_\rho(\delta) \rightarrow 0$ as $\delta \downarrow 0$, such that, for every finite or countable real family (b_j) with $\sum_j b_j^2 < \infty$,*

$$\left| \prod_j \phi(tb_j) - \exp\left(-\frac{t^2}{2} \sum_j b_j^2\right) \right| \leq \omega_\rho(\delta) t^2 \sum_j b_j^2 \quad \text{if} \quad \sup_j |tb_j| \leq \delta. \quad (47)$$

For a countable family, the product denotes the characteristic function of the L^2 sum $\sum_j b_j X_j$.

Proof. The centered second-moment assumption gives

$$\phi(u) = 1 - \frac{u^2}{2} + o(u^2).$$

Consequently

$$\omega_\rho(\delta) := \sup_{0 < |u| \leq \delta} \frac{|\phi(u) - e^{-u^2/2}|}{u^2} \longrightarrow 0, \quad \omega_\rho(0) := 0.$$

Both factors have modulus at most one. Telescoping finite products therefore bounds their difference by $\sum_j \omega_\rho(\delta) t^2 b_j^2$, proving (47). The countable case follows from L^2 convergence of the partial sums. $\square$

Lemma 3.8 (Qualitative rigidity of a unit row). *Let $a^{(n)} \in \ell^2$ satisfy $\|a^{(n)}\|_2 = 1$. If*

$$\text{Law}\left(\sum_j a_j^{(n)} X_j\right) \Longrightarrow \rho,$$

then

$$\inf_{\substack{j \geq 1, \varsigma \in \{-1, 1\} \\ \text{Law}(\varsigma X) = \rho}} \|a^{(n)} - \varsigma e_j\|_2 \longrightarrow 0. \quad (48)$$

Proof. To separate surviving coefficients from a diffuse tail, reorder each row by decreasing absolute value; this preserves the law of the sum and gives $|a_j^{(n)}| \leq j^{-1/2}$. From any subsequence, extract a further one with $a_j^{(n)} \rightarrow a_j$ for every j . Fatou gives

$$\tau_a := \sum_j a_j^2 \leq 1.$$

For fixed m, t , lemma 3.7 replaces the tail characteristic function by

$$\exp\left[-\frac{t^2}{2} \left(1 - \sum_{j \leq m} (a_j^{(n)})^2\right)\right]$$

with error at most $t^2\omega_\rho(|t|/\sqrt{m+1})$, uniformly in n . Letting first $n \rightarrow \infty$ and then $m \rightarrow \infty$ gives

$$e^{-(1-\tau_a)t^2/2} \prod_j \phi(a_j t). \quad (49)$$

Thus the limit is $\sum_j a_j X_j$ plus an independent $N(0, 1 - \tau_a)$ variable.

The limiting law is ρ . If every surviving coefficient is smaller than one, substituting this representation into itself repeatedly makes all coefficients small. The resulting law must then be Gaussian. To see this, suppose that no $|a_j|$ is one. If all a_j vanish, (49) already makes X Gaussian. Otherwise $m_a := \max_j |a_j|$ exists and lies in $(0, 1)$. For $\mathbf{j} = (j_1, \dots, j_k) \in \mathbb{N}^k$, set $b_{\mathbf{j}} := a_{j_1} \cdots a_{j_k}$. Iterating the characteristic-function identity (49) k times gives

$$\phi(t) = e^{-(1-\tau_a^k)t^2/2} \prod_{\mathbf{j} \in \mathbb{N}^k} \phi(tb_{\mathbf{j}}), \quad (50)$$

where

$$\sum_{\mathbf{j}} b_{\mathbf{j}}^2 = \tau_a^k, \quad \sup_{\mathbf{j}} |b_{\mathbf{j}}| \leq m_a^k.$$

The substitution is legitimate in L^2 at every finite depth. Applying lemma 3.7 to (50) yields

$$|\phi(t) - e^{-t^2/2}| \leq t^2 \tau_a^k \omega_\rho(|t|m_a^k) \rightarrow 0.$$

This contradicts non-Gaussianity. Hence some $a_j = \varsigma \in \{-1, 1\}$. All other coefficients and the Gaussian summand vanish, and the limiting identity gives $\text{Law}(\varsigma X) = \rho$.

If (48) failed, a subsequence would stay a fixed distance from every allowed signed unit vector. The preceding extraction produces an allowed ςe_j with

$$\|a^{(n)} - \varsigma e_j\|_2^2 = 2 - 2\varsigma a_j^{(n)} \rightarrow 0.$$

Undoing the permutations is a contradiction. $\square$

An absolutely continuous change of a product law can be approximated by changing only a finite prefix. Beyond that prefix, coordinate permutations are almost invisible and the marginals are almost unchanged. We will use both consequences of this observation.

Lemma 3.9 (Uniform invariance under tail symmetries). *Let μ be the i.i.d. product law and let $\nu \ll \mu$. Let $\mathcal{T}_N$ consist of the members of $\mathcal{P}$ that change only finitely many coordinates and fix coordinates $1, \dots, N$. Then*

$$\sup_{S \in \mathcal{T}_N} h(\nu, S_{\#}\nu) \rightarrow 0. \quad (51)$$

Also, the one-coordinate marginals of ν converge to the common coordinate law of μ , uniformly over coordinates greater than N .

Proof. Put $\ell = d\nu/d\mu$ and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. The martingale convergence theorem gives $\|\ell - \ell_N\|_1 \rightarrow 0$. Every $S \in \mathcal{T}_N$ preserves μ and satisfies $\ell_N \circ S^{-1} = \ell_N$. Thus

$$h(\nu, S_{\#}\nu)^2 \leq \|\ell - \ell \circ S^{-1}\|_{L^1(\mu)} \leq 2\|\ell - \ell_N\|_{L^1(\mu)},$$

which proves (51). For $i > N$, the marginal density is $\mathbb{E}_\mu[\ell \mid x_i]$, while independence gives $\mathbb{E}_\mu[\ell_N \mid x_i] = 1$, so

$$\|\mathbb{E}_\mu[\ell \mid x_i] - 1\|_{L^1(\mu)} \leq \|\ell - \ell_N\|_{L^1(\mu)}.$$

Taking the supremum over $i > N$ proves the marginal assertion. $\square$

The marginal conclusion identifies the limiting row laws. To use unit-row rigidity, we also need their lengths. In fact, absolute continuity controls the covariance on the entire remote subspace, giving a compact covariance defect. The only obstacle is that covariance is unbounded; truncating the input variables will justify passing from closeness of laws to closeness of second moments.

Lemma 3.10 (Compactness of the covariance defect). *Let T be bounded on ℓ^2 . If $\nu_T \ll \mu$, then*

$$TT^* - I \text{ is compact.}$$

*If T is boundedly invertible, $T^*T - I$ is compact as well.*

Proof. We first show that, for every $B < \infty$, the squares of

$$Y_b := \sum_j b_j X_j, \quad \|b\|_2 \leq B, \quad (52)$$

are uniformly integrable. Let

$$X^{(M)} = X \mathbf{1}_{\{|X| \leq M\}} - \mathbb{E}[X \mathbf{1}_{\{|X| \leq M\}}], \quad Y_b^{(M)} = \sum_j b_j X_j^{(M)}.$$

The variables $X^{(M)}$ are bounded and centered, and converge to X in L^2 . Independence gives

$$\sup_{\|b\|_2 \leq B} \|Y_b - Y_b^{(M)}\|_2 \leq B \|X - X^{(M)}\|_2 \longrightarrow 0.$$

For fixed M , the fourth-moment formula for independent centered variables gives

$$\sup_{\|b\|_2 \leq B} \mathbb{E}|Y_b^{(M)}|^4 < \infty.$$

It also makes the finite partial sums Cauchy in L^4 , so the bound holds for countable sums. Moreover,

$$\sup_{\|b\|_2 \leq B} \|Y_b^2 - (Y_b^{(M)})^2\|_1 \leq B^2 \|X - X^{(M)}\|_2 (\|X\|_2 + \|X^{(M)}\|_2) \longrightarrow 0.$$

Thus Y_b^2 is uniformly approximated in L^1 by a uniformly integrable family, proving the claim using only the second moment of X .

Write $\nu = \nu_T$, $\ell = d\nu/d\mu$, and $\ell_N = \mathbb{E}_\mu[\ell \mid \sigma(x_1, \dots, x_N)]$. For a finitely supported unit vector a supported on coordinates greater than N , put $F_a(x) = (\sum_i a_i x_i)^2$. Under μ , these are squares from (52); under ν , they are the squares of Y_{T^*a} under μ . Since $\|T^*a\|_2 \leq \|T\|_{\text{op}}$, they are uniformly integrable under both laws. The bounded statistic $F_a \wedge L$ depends only on tail coordinates, so independence gives

$$|\mathbb{E}_\nu(F_a \wedge L) - \mathbb{E}_\mu(F_a \wedge L)| \leq L \|\ell - \ell_N\|_1.$$

Letting first $N \rightarrow \infty$ and then $L \rightarrow \infty$, uniformly in a , gives

$$\sup_{\substack{\|a\|_2=1, \text{ } a \text{ finitely supported} \\ \text{supp } a \subset \{N+1, N+2, \dots\}}} |\langle (TT^* - I)a, a \rangle| \longrightarrow 0.$$

Since $TT^* - I$ is self-adjoint, density of finite vectors in the tail subspace implies

$$\|(I - P_N)(TT^* - I)(I - P_N)\|_{\text{op}} \longrightarrow 0.$$

The complementary rows and columns are finite rank, so $TT^* - I$ is compact. If T is invertible, then $T^*T - I = T^{-1}(TT^* - I)T$. $\square$

The axes are now identifiable, but the errors could still tend to zero too slowly to be square summable. To exclude this, compare a finite group of imperfect matches with an equal number of much better matches farther out. Exchange the two groups at both input and output. Exact matches would make the input exchange followed by T agree with T followed by the output exchange. The discrepancy therefore measures the matching error. Composing with the inverse maps turns it into a perturbation of the identity, where the local Hellinger lower bound applies. Absolute continuity, meanwhile, makes the remote output exchange invisible. The next proof makes these two comparisons quantitative.

Lemma 3.11 (Hilbert–Schmidt necessity under a second-moment hypothesis). *Assume (1). If $T \in \text{GL}(H)$ and $\nu_T \sim \mu$, then*

$$T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}. \quad (53)$$

When T is orthogonal, the condition $xg' \in L^2$ can be omitted.

Proof. By lemma 3.10, $\|T^*e_i\|_2 \rightarrow 1$, and by lemma 3.9, the law of the corresponding row sum tends to ρ . Normalization changes that sum by $o_{L^2}(1)$, so lemma 3.8 gives indices $\iota(i)$ and ρ -preserving signs ς_i such that

$$\widehat{e}_i := \varsigma_i e_{\iota(i)}, \quad \|T^*e_i - \widehat{e}_i\|_2^2 \longrightarrow 0. \quad (54)$$

The indices $\iota(i)$ are eventually distinct: if $\iota(i) = \iota(j)$, then $\varsigma_i T^*e_i - \varsigma_j T^*e_j$ has arbitrarily small norm by (54), whereas bounded invertibility gives the lower bound $\sqrt{2}/\|T^{-1}\|_{\text{op}}$. Fix N_0 so that ι is injective on $i > N_0$.

Since $\iota(i)$ leaves every finite set, compactness of $T^*T - I$ gives

$$\|T\widehat{e}_i\|_2^2 \longrightarrow 1.$$

Together with $\langle T\widehat{e}_i, e_i \rangle = \langle \widehat{e}_i, T^*e_i \rangle \rightarrow 1$, this yields

$$\text{err}_i := \|T^*e_i - \widehat{e}_i\|_2^2 + \|T\widehat{e}_i - e_i\|_2^2 \longrightarrow 0. \quad (55)$$

We have matched both the row and its corresponding column; together their errors will control the discrepancy between the exchanges. We claim that $\sum_{i>N_0} \text{err}_i < \infty$. Choose the local constants in lemma 3.6 so that

$$h(\mu, \nu_W) \geq c_{\text{HS}} \|W - I\|_{\mathcal{S}_2} \quad \text{when } W \in \text{GL}(H), \quad \|W - I\|_{\mathcal{S}_2} \leq \delta_{\text{HS}}. \quad (56)$$

Suppose the claim fails, and choose $\delta > 0$ so that $3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}$. In an arbitrarily remote tail where $\text{err}_i \leq \delta^2$, choose a finite set E satisfying

$$\delta^2 \leq \sum_{i \in E} \text{err}_i \leq 2\delta^2. \quad (57)$$

Pair each $i \in E$ with a distinct later index k_i such that, for $G = \{k_i : i \in E\}$, the sum $\sum_{k \in G} \text{err}_k$ is arbitrarily small.

Let S exchange each output pair (e_i, e_{k_i}) , and let P exchange the aligned signed input pairs $(\hat{e}_i, \hat{e}_{k_i})$, acting identically on their orthogonal complements. Both are involutions in $\mathcal{P}$, so they preserve μ . Define

$$W = PT^{-1}ST.$$

Then

$$W - I = PT^{-1}(ST - TP), \quad \frac{\|ST - TP\|_{\mathcal{S}_2}}{\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq \|T^{-1}\|_{\text{op}} \|ST - TP\|_{\mathcal{S}_2}. \quad (58)$$

Hellinger invariance and $P_{\#}\mu = \mu$ give

$$h(\mu, \nu_W) = h(\nu_T, S_{\#}\nu_T). \quad (59)$$

All differences have finite rank, and [lemma 3.1](#) supplies the common measurable carrier needed for this identity.

The norm comparison becomes transparent if the auxiliary matches are exact. We can make this change at arbitrarily small Hilbert–Schmidt cost. Define

$$T_0^* e_k = \hat{e}_k, \quad T_0 \hat{e}_k = e_k \quad (k \in G).$$

On the complementary block let T_0 agree with T . Orthogonal block decomposition gives

$$\|T - T_0\|_{\mathcal{S}_2}^2 \leq \sum_{k \in G} (\|T^* e_k - \hat{e}_k\|_2^2 + \|T \hat{e}_k - e_k\|_2^2) = \sum_{k \in G} \text{err}_k. \quad (60)$$

Using the decompositions $E \oplus (E \cup G)^c \oplus G$ on the output and the corresponding aligned input coordinates, write

$$T_0 = \begin{pmatrix} A & B & 0 \\ C & D & 0 \\ 0 & 0 & I \end{pmatrix}, \quad ST_0 - T_0 P = \begin{pmatrix} 0 & -B & I - A \\ C & 0 & -C \\ A - I & B & 0 \end{pmatrix}.$$

Here B, C are Hilbert–Schmidt, and the error on E is

$$\sum_{i \in E} (\|T_0^* e_i - \hat{e}_i\|_2^2 + \|T_0 \hat{e}_i - e_i\|_2^2) = 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2,$$

while the discrepancy satisfies

$$\|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 = 2(\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2),$$

Therefore,

$$\begin{aligned} 2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2 &\leq \|ST_0 - T_0 P\|_{\mathcal{S}_2}^2 \\ &\leq 2(2\|A - I\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2). \end{aligned} \quad (61)$$

As $\sum_{k \in G} \text{err}_k \rightarrow 0$, the first displayed error tends to $\sum_{i \in E} \text{err}_i$, while

$$\|(ST - TP) - (ST_0 - T_0 P)\|_{\mathcal{S}_2} \leq 2\|T - T_0\|_{\mathcal{S}_2} \rightarrow 0.$$

Hence (57) and (61) allow the auxiliary indices to be chosen so that

$$\frac{\delta}{2} \leq \|ST - TP\|_{\mathcal{S}_2} \leq 3\delta.$$

It follows from (58) that

$$\frac{\delta}{2\|T\|_{\text{op}}} \leq \|W - I\|_{\mathcal{S}_2} \leq 3\|T^{-1}\|_{\text{op}}\delta < \delta_{\text{HS}}.$$

By (56), $h(\mu, \nu_W) \geq c_{\text{HS}}\delta/(2\|T\|_{\text{op}})$. But S is supported in the chosen remote tail, so lemma 3.9 and (59) make this quantity tend to zero as the tail recedes. This contradiction proves $\sum_{i>N_0} \text{err}_i < \infty$.

It remains to complete the tail matching to a permutation of all coordinates. The number of unused inputs must equal the number of unmatched outputs; this follows from the Fredholm index. Define Q_0 by the matched rows $\hat{e}_i$ after the fixed initial cutoff N_0 , and by zero rows for $i \leq N_0$. The summability just proved implies

$$\begin{aligned} \|T - Q_0\|_{\mathcal{S}_2}^2 &= \sum_{i \leq N_0} \|T^* e_i\|_2^2 + \sum_{i > N_0} \|T^* e_i - \hat{e}_i\|_2^2 \\ &\leq \sum_{i \leq N_0} \|T^* e_i\|_2^2 + \sum_{i > N_0} \text{err}_i < \infty. \end{aligned}$$

Thus Q_0 is a compact perturbation of the invertible operator T , hence Fredholm of index zero. Its cokernel is the span of the first N_0 output coordinates and its kernel is spanned by the unused input coordinates, so exactly N_0 columns are unused. Matching them to the first N_0 rows with positive signs completes Q_0 to $Q \in \mathcal{P}$ and proves (53).

If T is orthogonal, both covariance defects vanish, and every $W = PT^*ST$ above is orthogonal. Thus only the orthogonal part of lemma 3.6 is used, and it requires no scale Fisher information. $\square$

Exact linear stabilizer. Exact invariance requires only the second moment.

Proof of corollary 1.2. Every member of $\mathcal{P}$ preserves the product law. Conversely, if $\nu_T = \mu$, covariance gives $TT^* = I$. Since T is boundedly invertible, it is orthogonal. Each of its unit rows gives a linear combination with the same law as X . Applying lemma 3.8 to the constant sequence consisting of this row shows that the row is an allowed signed unit vector. Orthogonality makes the selected coordinates a bijection. Thus $T \in \mathcal{P}$. $\square$

Proof of theorem 1.1. Necessity follows from lemma 3.11. If $T - Q \in \mathcal{S}_2$ for $Q \in \mathcal{P}$, then $TQ^{-1} = I + (T - Q)Q^{-1}$ is invertible and differs from the identity by a Hilbert–Schmidt operator. By proposition 3.4, $\nu_{TQ^{-1}} \sim \mu$. Since $Q_{\#}\mu = \mu$, this is $\nu_T \sim \mu$. For $K = TQ^{-1} - I$, the same proposition shows that the likelihood ratios for $I + P_n K P_n$ converge to $d\nu_T/d\mu$ in $L^1(\mu)$, and their square roots converge in $L^2(\mu)$. Lemmas 3.2 and 3.3 give the projected-energy characterization, the singular alternative, and two-sided uniform integrability. The exact linear stabilizer was proved above. $\square$

Proof of corollary 1.3. Necessity is lemma 3.11. For sufficiency, write $UQ^{-1} = I + (U - Q)Q^{-1}$. This is orthogonal and differs from the identity by a Hilbert–Schmidt operator. Apply proposition 3.5, and then use $Q_{\#}\mu = \mu$. The alternative follows from lemma 3.2. $\square$

3.6 Returning to the global finite-dimensional bound

The infinite-dimensional theorem also answers the question left open at the end of Section 2. Suppose matrices could stay a fixed Frobenius distance from every coordinate symmetry while their Hellinger distances tended to zero. Choose them with summable Hellinger losses and stack them in independent blocks. Kakutani would give equivalence, whereas our classification would force those same blocks to approach coordinate symmetries. This contradiction gives uniform separation in every dimension.

Proof of theorem 2.7. Fix $0 < a \leq b < \infty$. We first prove uniform separation away from the exact linear stabilizer: for every $\varepsilon > 0$,

$$\inf \left\{ h(\mu_n, A_{\#}\mu_n)^2 : \begin{array}{l} n \geq 1, A \in \text{GL}(n, \mathbb{R}), \\ s_{\min}(A) \geq a, s_{\max}(A) \leq b, \\ d_n(A) \geq \varepsilon \end{array} \right\} > 0. \quad (62)$$

If this fails, choose matrices $A_k \in \text{GL}(n_k, \mathbb{R})$ in the displayed class whose Hellinger distances tend to zero. Since

$$-2 \log A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) = -2 \log \left(1 - \frac{1}{2} h(\mu_{n_k}, (A_k)_{\#}\mu_{n_k})^2 \right),$$

we may pass to a subsequence for which the sum of these block energies is finite.

On consecutive coordinate blocks define $A = \bigoplus_{k \geq 1} A_k \in \text{GL}(H)$. Each block law is equivalent to μ_{n_k} , and the energy summability gives

$$\prod_{k \geq 1} A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) > 0.$$

Kakutani's theorem [17] gives $\nu_A \sim \mu$, so theorem 1.1 supplies $Q \in \mathcal{P}$ with

$$D := A - Q \in \mathcal{S}_2(H). \quad (63)$$

Let Π_k be the orthogonal projection onto the k -th output block. The orthogonality of these blocks gives

$$\sum_k \|\Pi_k D\|_{\mathcal{S}_2}^2 = \|D\|_{\mathcal{S}_2}^2 < \infty, \quad \text{hence} \quad \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0.$$

If e_i belongs to the k th output block but Q^*e_i lies outside the corresponding input block, then the two vectors have disjoint supports and

$$\|A^*e_i - Q^*e_i\|_2^2 = \|A_k^*e_i\|_2^2 + 1 \geq a^2 + 1. \quad (64)$$

For large k this contradicts $\|\Pi_k D\|_{\mathcal{S}_2}^2 < a^2 + 1$. Thus Q^* preserves every sufficiently large block; its restriction is a bijection and defines $Q_k \in \mathcal{P}_{n_k}$. Hence

$$d_{n_k}(A_k) \leq \|A_k - Q_k\|_F = \|\Pi_k D\|_{\mathcal{S}_2} \longrightarrow 0,$$

contrary to $d_{n_k}(A_k) \geq \varepsilon$. This proves (62).

Let $c_{\text{loc}}, C_{\text{loc}}, \delta_{\text{loc}}$ be the local constants, with $0 < \delta_{\text{loc}} < 1$. Choose a minimizer $Q_n \in \mathcal{P}_n$ for $d_n(T)$. Since $(Q_n)_{\#}\mu_n = \mu_n$,

$$h(\mu_n, T_{\#}\mu_n) = h(\mu_n, (TQ_n^{-1})_{\#}\mu_n), \quad \|TQ_n^{-1} - I\|_F = d_n(T).$$

For $d_n(T) \leq \delta_{\text{loc}}$, the local theorem gives both bounds. For $d_n(T) > \delta_{\text{loc}}$, (62) gives the lower bound and $h^2 \leq 2$ the upper bound. Taking the smaller lower constant and $C = \max\{C_{\text{loc}}^2, 2\delta_{\text{loc}}^{-2}\}$ proves (33). $\square$

4 Symmetric Stable Products

The second moment enters the preceding classification through the cost of mixing one coordinate into another. A shear $(x, y) \mapsto (x + ty, y)$ moves the first coordinate by order one when $|y|$ is of order $|t|^{-1}$. For a symmetric α -stable input, this event has probability of order $|t|^\alpha$. It changes the local mixing cost from quadratic to α -th order, even though the density is smooth and its Fisher information is finite.

When one input moves several outputs, all those displacements occur together. Their size is the ℓ^2 -length of the corresponding matrix column. This suggests a sum of column lengths to the power α , while individual changes of scale keep their quadratic cost. We prove that this is exactly the geometry of stable linear equivalence. The distinction from coordinatewise affine transformations [20, 23, 26] is the dependence created by a shared input. Fix $0 < \alpha < 2$, let ρ_α be the standard symmetric α -stable law, and put

$$\int_{\mathbb{R}} e^{itx} \rho_\alpha(dx) = e^{-|t|^\alpha}, \quad \mu_\alpha = \rho_\alpha^{\otimes \mathbb{N}}, \quad \mu_{\alpha,n} = \rho_\alpha^{\otimes n}. \quad (65)$$

For this symmetric law, $\mathcal{P}$ is the full signed-permutation group and $\mathcal{P}_n$ its n -dimensional version. Let $(X_j)_{j \geq 1}$ be independent with common law ρ_α , and write $\mathbf{X} = (X_j)_{j \geq 1}$. We use the symmetric unit-scale convention of [25]. The density f_α is smooth and positive and has finite location and scale Fisher information despite its infinite second moment [4]: the density and derivative asymptotics in [40] give $f_\alpha(x) \asymp (1 + |x|)^{-1-\alpha}$ and location score $O((1 + |x|)^{-1})$. The normalization fixes the location parameter at zero, including when $\alpha \leq 1$ and the mean does not exist.

The tails also determine which rows define random variables. A stable series $\sum_j a_j X_j$ converges almost surely exactly for $a \in \ell^\alpha$ [30]. For a matrix T with such rows, let $\Phi_{\alpha,T}$ be its row-series map and $\nu_{\alpha,T} = (\Phi_{\alpha,T})_\# \mu_\alpha$. The realization is *compatible* if some matrix S , also with ℓ^α rows, has measurable row-series maps satisfying

$$\Phi_{\alpha,S} \circ \Phi_{\alpha,T} = I, \quad \Phi_{\alpha,T} \circ \Phi_{\alpha,S} = I$$

on a common full-measure carrier for the corresponding orbit laws.

For a matrix K , put $F_\alpha(K) = \sum_j \|K e_j\|_2^\alpha$. For $M = (m_{ij})$, define

$$\mathfrak{e}_\alpha(M) = \sum_i |m_{ii} - 1|^2 + F_\alpha(M - \text{diag } M). \quad (66)$$

The two terms measure the scale changes and the simultaneous displacements caused by each input. As before, we compare the first n output coordinates through their Hellinger energy:

$$E_{\alpha,n}(T) = -2 \log A(\mu_{\alpha,n}, (\pi_n)_\# \nu_{\alpha,T}). \quad (67)$$

Theorem 4.1 (Complete criterion for symmetric stable coordinates). *Let $0 < \alpha < 2$ and suppose that T admits a compatible stable row-series realization. Then*

$$\boxed{\begin{aligned} \nu_{\alpha,T} \sim \mu_\alpha &\iff T|_H \in \text{GL}(H) \text{ and } \mathfrak{e}_\alpha(TQ^{-1}) < \infty \text{ for some } Q \in \mathcal{P} \\ &\iff \sup_n E_{\alpha,n}(T) < \infty, \end{aligned}} \quad (68)$$

If these conditions fail, then $\nu_{\alpha,T} \perp \mu_\alpha$ and $E_{\alpha,n}(T) \uparrow \infty$. For $T \in \text{GL}(H)$, the displayed summability condition itself supplies compatible stable row-series realizations of T and T^{-1} . The exact linear stabilizer in this class is $\mathcal{P}$.

The coordinate symmetries are unchanged, but their neighbourhood of equivalent mixings is smaller in the off-diagonal directions. For instance, a square-summable sequence of independent shears can now give a singular image. At the end of the section we contrast this with a single input spread over infinitely many outputs.

Proof. Two ways of viewing a stable law are useful for different parts of the proof. As a Gaussian scale mixture it gives the upper bound. After Fourier transformation of its square-root density, its finite Fisher information becomes a finite second moment, allowing us to recover coordinate matching from Section 3. Finally, the large-input events described above detect the additional column cost that the second-moment argument cannot see.

Translations determine the operator domain. Put $g_\alpha = \sqrt{f_\alpha}$ and define

$$a_\alpha(u) = \int_{\mathbb{R}} g_\alpha(x) g_\alpha(x - u) dx.$$

Finite positive location Fisher information and decay of translations in L^2 give

$$1 - a_\alpha(u) \asymp_\alpha \min\{u^2, 1\}. \quad (69)$$

Kakutani's theorem therefore identifies ℓ^2 as the translation space of μ_α . Under equivalence, conjugating translations through the compatible realization shows that T and its inverse matrix S preserve ℓ^2 ; the closed graph theorem gives $T|_H \in \text{GL}(H)$.

For the dichotomy itself, the two laws need not have the same translation space. The product is quasi-invariant and ergodic under c_{00} ; its image has the same properties under $T(c_{00})$, by translating the inputs. Okazaki's theorem [27], which allows different linear subspaces, therefore gives equivalence or singularity. The compatible inverse makes every finite family of output rows linearly independent. A nonsingular input minor then gives a positive output density, as in lemma 3.3. Passing to the limiting affinity yields

$$\nu_{\alpha,T} \sim \mu_\alpha \iff \sup_n E_{\alpha,n}(T) < \infty, \quad (70)$$

with singularity and divergent energy on the other branch.

Averaging the Gaussian comparison. The column cost behaves well under the operations needed for approximation:

$$\begin{aligned} F_\alpha(AK) &\leq \|A\|_{\text{op}}^\alpha F_\alpha(K), & F_\alpha(KD) &\leq \|D\|_{\text{op}}^\alpha F_\alpha(K), \\ \|K^* \gamma\|_\alpha^\alpha &\leq F_\alpha(K) \|\gamma\|_2^\alpha, \end{aligned} \quad (71)$$

for bounded A , bounded diagonal D , and $\gamma \in \ell^2$. In particular, finite column cost gives admissible stable rows.

To bound the change of law, condition on the random scales in a Gaussian mixture representation. We can then use the Gaussian affinity formula, and averaging over the scales produces the column exponent α . The resulting estimate is

$$h^2(A_{\#} \mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_\alpha \min\{\mathfrak{e}_\alpha(A), 1\} \quad (72)$$

for every invertible real $n \times n$ matrix A , with a constant independent of n . Put $\beta = \alpha/2$ and use the positive-stable representation $X_i = \sqrt{2V_i} Z_i$, where $\mathbb{E} e^{-tV_i} = e^{-t^\beta}$ and the Z_i are independent

standard Gaussians [30, 31]. Conditioning on $V = \text{diag}(V_i)$, Hellinger data processing and the Gaussian affinity formula bound the left side of (72) by a constant times

$$\mathbb{E} \min\{\|V^{-1/2}(A - I)V^{1/2}\|_F^2, 1\}.$$

For $c_j^2 = \sum_{i \neq j} |a_{ij}|^2$, source column j contributes $W_j = V_j \sum_{i \neq j} |a_{ij}|^2 / V_i$. The positive-stable tail and truncated-moment bounds, together with $\mathbb{E}V_1^{-1} < \infty$, give

$$\mathbb{P}(c_j^2 V_j > 1) + \mathbb{E}[W_j; c_j^2 V_j \leq 1] \leq C_\alpha c_j^{2\beta} = C_\alpha c_j^\alpha.$$

Summation over j , followed by the quadratic diagonal-scale estimate, proves (72).

With this estimate, finite-coordinate approximation follows the same principle as in Section 3. Remove the measure-preserving Q and write $M = D + K$, where $D = \text{diag } M$ has entries d_i , $(d_i - 1) \in \ell^2$, and $F_\alpha(K) < \infty$. Only finitely many d_i are nonpositive. Replace each of them by 1 in D and add $d_i - 1$ to the corresponding diagonal entry of K ; this preserves both summability conditions and makes D positive and boundedly invertible. Set

$$D_n = I + P_n(D - I)P_n, \quad K_n = P_n K P_n, \quad M_n = D_n + K_n,$$

so that $M_n \rightarrow M$ in operator norm and the M_n are eventually invertible with uniformly bounded inverses. The identity $M_n^{-1} = D_n^{-1} - M_n^{-1} K_n D_n^{-1}$ and (71) give these inverses uniformly bounded diagonal parts and column costs. Applied to $M_n^{-1}(M_m - M_n)$, the same bounds show that its off-diagonal column cost and diagonal square cost tend to zero; here $F_\alpha(K_m - K_n) \rightarrow 0$, $\|D_m - D_n\|_{\text{op}} \rightarrow 0$, and finite column cost implies Hilbert–Schmidt membership. Hence (72) makes the square-root likelihoods of ν_{α, M_n} Cauchy in L^2 . The limit has norm one, and ℓ^α -convergence of every row identifies its square with $d\nu_{\alpha, M}/d\mu_\alpha$. This proves absolute continuity, which (70) upgrades to equivalence. The identity $(D + K)^{-1} = D^{-1} - (D + K)^{-1} K D^{-1}$ gives the same admissible form for the inverse. Continuity of stable random functionals from ℓ^α to convergence in probability then passes finite-matrix composition to the row-series limits and supplies compatible versions.

Recovering coordinate matching through Fourier transformation. Now suppose $\nu_{\alpha, T} \sim \mu_\alpha$. The lack of a second moment prevents us from applying the row-rigidity argument to the stable variables themselves. Plancherel suggests a way around this: the square-integrable derivative of g_α becomes a square-integrable frequency weight. Thus Fourier transformation gives an auxiliary probability law with a finite second moment. Its characteristic function is the translation affinity a_α , the adjoint-measure construction of [34].

Let $\widehat{g}_\alpha$ be the unitary Fourier transform of g_α , and set

$$\rho_\alpha^\sharp(d\xi) = |\widehat{g}_\alpha(\xi)|^2 d\xi.$$

The law $\rho_\alpha^\sharp$ is symmetric, has variance $\|g'_\alpha\|_2^2 \in (0, \infty)$, and has characteristic function a_α . Since $\widehat{g}_\alpha * \widehat{g}_\alpha$ is a positive multiple of $e^{-|t|^\alpha}$, we also have $|\widehat{g}_\alpha|^2 * |\widehat{g}_\alpha|^2 > 0$ almost everywhere. To apply the finite-variance classification, we smooth this auxiliary law by convolution. Polynomial stable tails give two-sided polynomial bounds for a_α , and a'_α is bounded. Choose an integer r so large that

$$\int_{\mathbb{R}} t^2 (|a_\alpha(t)|^{2r} + |(a'_\alpha)^r(t)|^2) dt < \infty.$$

Fourier inversion gives $(\rho_\alpha^\#)^{*r}$ a bounded-variation density, whose third convolution has finite location Fisher information by the smoothing estimate in [4]. If f_{3r} is the density of $(\rho_\alpha^\#)^{*3r}$, U_1, U_2 are independent with this density, and s_{3r}, s_{6r} are the scores of f_{3r} and $f_{3r} * f_{3r}$, then

$$xs_{6r}(x) - 1 = 2\mathbb{E}[U_1 s_{3r}(U_2) \mid U_1 + U_2 = x].$$

Thus finite variance and location Fisher information give finite scale Fisher information after one further convolution. Set $k = 6r$ and $\mu_{\alpha,k}^\# = ((\rho_\alpha^\#)^{*k})^{\otimes \mathbb{N}}$. After rescaling to variance one, $(\rho_\alpha^\#)^{*k}$ has an a.e.-positive density satisfying (1); it is non-Gaussian because its characteristic function a_α^k has a polynomial lower bound.

Let $f_{\alpha,T,n}$ be the density of the first n output marginal and put

$$\ell_{\alpha,T,n} = \frac{f_{\alpha,T,n}}{f_\alpha^{\otimes n}}, \quad \ell_{\alpha,T} = \frac{d\nu_{\alpha,T}}{d\mu_\alpha}.$$

Here $\ell_{\alpha,T,n}$ is lifted to the product space and $\sqrt{f_{\alpha,T,n}} = g_\alpha^{\otimes n} \sqrt{\ell_{\alpha,T,n}}$. Equivalence gives $\sqrt{\ell_{\alpha,T,n}} \rightarrow \sqrt{\ell_{\alpha,T}}$ in $L^2(\mu_\alpha)$.

We must also transfer the change of measure to the auxiliary product. Fourier transformation preserves the L^2 convergence of square roots, and taking squared moduli converts it to total-variation convergence of probability laws. To obtain the convolution power just constructed, take k tensor copies of $\sqrt{f_{\alpha,T,n}}$, Fourier transform, square the modulus, and add the k frequencies at each site. Let $\Xi_{\alpha,T,n}$ be the product of the resulting law on $\mathbb{R}^n$ with $((\rho_\alpha^\#)^{*k})^{\otimes \{i>n\}}$. The first factor has a Lebesgue density and the tail density is positive almost everywhere; hence $\Xi_{\alpha,T,n} \ll \mu_{\alpha,k}^\#$.

For $m > n$, pad $\sqrt{f_{\alpha,T,n}}$ by $m - n$ factors of g_α . Unitarity, contraction under pushforward, and $\| |u|^2 - |v|^2 \|_1 \leq 2\|u - v\|_2$ give

$$\|\Xi_{\alpha,T,m} - \Xi_{\alpha,T,n}\|_{\text{TV}} \leq 2\|(\sqrt{\ell_{\alpha,T,m}})^{\otimes k} - (\sqrt{\ell_{\alpha,T,n}})^{\otimes k}\|_{L^2(\mu_\alpha^{\otimes k})} \leq 2k\|\sqrt{\ell_{\alpha,T,m}} - \sqrt{\ell_{\alpha,T,n}}\|_{L^2(\mu_\alpha)}.$$

Hence $\Xi_{\alpha,T,n}$ converges in total variation to a probability $\Xi_{\alpha,T} \ll \mu_{\alpha,k}^\#$.

Let $\nu_{\alpha,T}^{[n]} = \ell_{\alpha,T,n} \mu_\alpha$. For every finitely supported z , Fourier autocorrelation and square-root convergence give

$$\begin{aligned} \widehat{\Xi}_{\alpha,T}(z) &= \lim_n \mathbf{A}(\nu_{\alpha,T}^{[n]}, (x \mapsto x + z)_{\#} \nu_{\alpha,T}^{[n]})^k \\ &= \mathbf{A}(\nu_{\alpha,T}, (x \mapsto x + z)_{\#} \nu_{\alpha,T})^k \\ &= \prod_i a_\alpha((T^{-1}z)_i)^k. \end{aligned}$$

The last expression is the characteristic function of $(\Phi_{(T^{-1})^*})_{\#} \mu_{\alpha,k}^\#$; agreement for finitely supported z identifies all finite marginals. Hence

$$(\Phi_{(T^{-1})^*})_{\#} \mu_{\alpha,k}^\# \ll \mu_{\alpha,k}^\#. \quad (73)$$

The dichotomy and classification in [theorem 1.1](#) upgrade this to equivalence and give $Q \in \mathcal{P}$ such that $(T^{-1})^* - Q \in \mathcal{S}_2$. Since

$$T - Q = -T((T^{-1})^* - Q)^* Q,$$

we obtain $T - Q \in \mathcal{S}_2$. After removing Q , write

$$TQ^{-1} = I + K, \quad K \in \mathcal{S}_2. \quad (74)$$

Detecting one large input. We have found the coordinate symmetry and obtained square-summable errors. To recover the stronger column cost, we look at a different scale. An input of size $\|Be_j\|_2^{-1}$ creates an order-one displacement in the other coordinates. Its stable probability is of order $\|Be_j\|_2^\alpha$, precisely the contribution of that column. The test below detects the direction of this displacement on the large-input event. When the total cost is small, simultaneous large inputs are rare enough that their contributions do not obscure one another. We prove that there are constants $\varepsilon_\alpha, c_\alpha > 0$ such that, for every n and every $B \in \mathbb{R}^{n \times n}$,

$$\text{diag } B = 0, \quad F_\alpha(B) \leq \varepsilon_\alpha \implies h^2((I + B)_\# \mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha F_\alpha(B). \quad (75)$$

To prove it, write $F = F_\alpha(B)$, $c_j = \|Be_j\|_2$, and $u_j = Be_j/c_j$ when $c_j \neq 0$. Then $\|u_j\|_2 = 1$, $(u_j)_j = 0$, and the stable scale contributed to output row i satisfies

$$\omega_i(B) := \sum_j |b_{ij}|^\alpha \leq F. \quad (76)$$

For a finite family of Hilbert-space vectors v_j , excluding the events $\{\|v_j\|_2 |X_j| > t\}$ and applying Chebyshev to the centered truncated sum gives a tail of order $t^{-\alpha} \sum_j \|v_j\|_2^\alpha$. Integration yields a modulus $\tau_\alpha(s) \downarrow 0$ such that

$$\mathbb{E} \min \left\{ \left\| \sum_j v_j X_j \right\|_2, 1 \right\} \leq \tau_\alpha \left(\sum_j \|v_j\|_2^\alpha \right). \quad (77)$$

We need a bounded statistic that detects an order-one displacement even when it is spread across many coordinates. Applying $\tanh$ first makes the coordinate variances finite; projecting in the direction of the displacement retains a positive response. Set $\psi(x) = \tanh x$. There are a smooth odd increasing χ , bounded by one, and $c_\alpha > 0$ such that every unit vector u satisfies

$$\Theta_u(x) = \sum_i u_i \psi(x_i), \quad \mathcal{M}_u(t) = \mathbb{E} \chi(\Theta_u(\mathbf{X} + tu)), \quad \mathcal{M}_u(t) \geq c_\alpha \quad (1 \leq t \leq 2). \quad (78)$$

Indeed, $\Theta_u(\mathbf{X})$ has mean zero and variance $\text{Var}(\psi(X))$, while its shift by tu changes it by at most 2 for $|t| \leq 2$. A fixed compact interval therefore carries all but uniformly small mass. Choose χ' positive on that interval enlarged by two. The bound $\inf_{|\zeta| \leq 2} \mathbb{E} \psi'(X + \zeta) > 0$ then makes $\mathcal{M}'_u(t)$ uniformly positive after the discarded contribution is absorbed. Oddness gives $\mathcal{M}_u(0) = 0$, proving (78).

The second part of the test selects the input responsible for the displacement. Choose a nonzero smooth odd function w , supported where $1 \leq |t| \leq 2$, with $0 \leq w \leq 1$ on the positive half-line. Put

$$\Theta_j(x) = \sum_{i \neq j} (u_j)_i \psi(x_i),$$

and define the bounded test

$$\Psi_B(x) = \max \left\{ -1, \min \left\{ \sum_{j: c_j \neq 0} w(c_j x_j) \chi(\Theta_j(x)), 1 \right\} \right\}.$$

For all sufficiently small c_j , stable tail asymptotics give

$$\mathbb{P}(1 \leq |c_j X_j| \leq 2) \leq C_\alpha c_j^\alpha, \quad \mathbb{E}|w(c_j X_j)| \geq c_\alpha c_j^\alpha. \quad (79)$$

The events $\{w(c_j X_j) \neq 0\}$ are independent, and the probability that at least two occur is $O_\alpha(F^2)$. Before truncation each summand has mean zero because $w(c_j X_j)$ is independent of $\Theta_j(\mathbf{X})$; hence $|\mathbb{E}\Psi_B(\mathbf{X})| \leq C_\alpha F^2$.

Set $\mathbf{Y} = (I + B)\mathbf{X}$ and fix $0 < \delta < 1/4$. By (76),

$$\mathbb{P}\{\exists i : |c_i(B\mathbf{X})_i| > \delta\} \leq C_\alpha \delta^{-\alpha} \sum_i c_i^\alpha \omega_i(B) \leq C_\alpha \delta^{-\alpha} F^2.$$

On the complementary event, Lipschitz continuity of w and (79) show that replacing every factor $w(c_j Y_j)$ by $w(c_j X_j)$ changes the expectation by at most $C_\alpha(\delta F + \delta^{-\alpha} F^2)$; removing the outer truncation costs $O_\alpha(F^2)$. The remaining-coordinate perturbation $B(I - e_j \otimes e_j)\mathbf{X}$ is independent of X_j , and (77) shows that replacing $\Theta_j(\mathbf{Y})$ by $\Theta_j(\mathbf{X} + c_j X_j u_j)$ costs at most $C_\alpha F \tau_\alpha(F)$ after summation. Because $(u_j)_j = 0$, conditioning on X_j and using Equations (78) and (79) gives

$$\sum_j \mathbb{E}[w(c_j X_j) \chi(\Theta_j(\mathbf{X} + c_j X_j u_j))] \geq c_\alpha F.$$

Thus

$$\mathbb{E}\Psi_B(\mathbf{Y}) - \mathbb{E}\Psi_B(\mathbf{X}) \geq c_\alpha F - C_\alpha\{\delta F + (1 + \delta^{-\alpha})F^2 + F\tau_\alpha(F)\}.$$

Choose δ , then F , so that the error is at most half the main response. The test is nonzero only when at least one such event occurs, and the coordinates of $\mathbf{Y}$ have stable scales at most $(1 + F)^{1/\alpha}$; hence

$$\mathbb{E}\Psi_B(\mathbf{X})^2 + \mathbb{E}\Psi_B(\mathbf{Y})^2 \leq C_\alpha F.$$

Hellinger Cauchy–Schwarz now gives (75).

From local detection to summability. To use the lower bound, we must isolate a small amount of column cost from a possibly divergent total. Changing the signs of a finite set of coordinates does exactly this: conjugation cancels the entries within each side of the cut and doubles those crossing it. As in the finite-variance proof, equivalence makes remote sign changes invisible. We will choose a cut whose cost stays large enough for the local estimate to detect it.

Let $B \in \mathcal{S}_2$ have ℓ^α rows, zero diagonal, and $\|B\|_{\text{op}} \leq 1/10$. Suppose $I + B$ admits a compatible stable row-series realization and $\nu_{\alpha, I+B} \sim \mu_\alpha$. For a finite coordinate set $\mathcal{J}$, let $Q_{\mathcal{J}}$ change the signs in $\mathcal{J}$ and set

$$\Delta_{\mathcal{J}} = Q_{\mathcal{J}} B Q_{\mathcal{J}} - B, \quad A_{\mathcal{J}} = (I + B)^{-1} Q_{\mathcal{J}} (I + B) Q_{\mathcal{J}}.$$

The matrix $\Delta_{\mathcal{J}}$ consists of the entries crossing the cut. Since $\mathcal{J}$ is finite, $\Delta_{\mathcal{J}}$ has finite column cost. Applying (75) to $P_n \Delta_{\mathcal{J}} P_n$ and using the sufficiency approximation extends that bound to $\Delta_{\mathcal{J}}$.

Put

$$K_{\mathcal{J}} = (I + B)^{-1} \Delta_{\mathcal{J}}, \quad D_{\mathcal{J}} = I + \text{diag } K_{\mathcal{J}}, \quad \tilde{B}_{\mathcal{J}} = D_{\mathcal{J}}^{-1} (I + K_{\mathcal{J}}) - I.$$

Then $A_{\mathcal{J}} = I + K_{\mathcal{J}}$ and $\text{diag } \tilde{B}_{\mathcal{J}} = 0$. Since $\|(I + B)^{-1} - I\|_{\text{op}} \leq 1/9$, projection away from each diagonal entry compares the off-diagonal columns of $K_{\mathcal{J}}$ and $\Delta_{\mathcal{J}}$. For small $F_\alpha(\Delta_{\mathcal{J}})$, summing these columnwise comparisons gives

$$F_\alpha(\tilde{B}_{\mathcal{J}}) \asymp_\alpha F_\alpha(\Delta_{\mathcal{J}}), \quad \|\text{diag } K_{\mathcal{J}}\|_2 \leq \frac{1}{9} F_\alpha(\Delta_{\mathcal{J}})^{1/\alpha}.$$

The lower bound for $\tilde{B}_{\mathcal{J}}$, the diagonal scale upper bound for $D_{\mathcal{J}}$, and the Hellinger triangle inequality now give

$$F_{\alpha}(\Delta_{\mathcal{J}}) \leq \varepsilon_* \implies h(\nu_{\alpha, A_{\mathcal{J}}}, \mu_{\alpha}) \geq c_* F_{\alpha}(\Delta_{\mathcal{J}})^{1/2}. \quad (80)$$

Indeed, by (75), the two terms have respective orders $F_{\alpha}(\Delta_{\mathcal{J}})^{1/2}$ and $F_{\alpha}(\Delta_{\mathcal{J}})^{1/\alpha}$, and the latter is smaller for small cost because $\alpha < 2$.

The finite-prefix argument in lemma 3.9, applied to the square-root likelihood, is uniform over finite tail sign changes. Since each such change preserves μ_{α} , Hellinger invariance gives

$$\sup_{\substack{\mathcal{J} \subset \{N+1, N+2, \dots\} \\ |\mathcal{J}| < \infty}} h(\mu_{\alpha}, \nu_{\alpha, A_{\mathcal{J}}}) \longrightarrow 0. \quad (81)$$

It remains to find such a cut. A random cut retains, on average, a fixed fraction of the off-diagonal cost. Adding its coordinates one at a time then lets us stop at a prescribed small threshold. Suppose $F_{\alpha}(B) = \infty$. Deleting finitely many columns or rows changes the cost finitely, the latter because each row lies in ℓ^{α} ; thus every tail has infinite cost. For a finite tail set $\mathcal{I}_0$, include each coordinate independently in $\mathcal{J}$ with probability one half. Columnwise concavity of $t^{\alpha/2}$ gives

$$\mathbb{E}_{\mathcal{J}} F_{\alpha}(P_{\mathcal{I}_0} \Delta_{\mathcal{J}} P_{\mathcal{I}_0}) \geq 2^{\alpha-1} F_{\alpha}(P_{\mathcal{I}_0} B P_{\mathcal{I}_0}). \quad (82)$$

As $\mathcal{I}_0$ exhausts the tail, some finite cuts have arbitrarily large cost. Equivalence makes the remote one-coordinate output marginals converge to ρ_{α} , so $\omega_i(B) \rightarrow 0$ in the notation of (76); Hilbert–Schmidt membership also gives $c_i \rightarrow 0$. Hence

$$F_{\alpha}(\Delta_{\{i\}}) = 2^{\alpha}(c_i^{\alpha} + \omega_i(B)) \longrightarrow 0.$$

For $\alpha < 1$, $F_{\alpha}^{1/\alpha}$ is a quasi-norm and its α -th power F_{α} is subadditive; for $\alpha \geq 1$, $F_{\alpha}^{1/\alpha}$ is a norm. Since adding i to a cut changes $\Delta_{\mathcal{J}}$ by a sign conjugate of $\Delta_{\{i\}}$, the increments of F_{α} in the first case and of $F_{\alpha}^{1/\alpha}$ in the second tend uniformly to zero. Fix $0 < \varepsilon < \varepsilon_*/2$, add the elements of a large cut from (82), and stop at the first threshold crossing. In every sufficiently remote tail this produces $\mathcal{J}$ with $\varepsilon \leq F_{\alpha}(\Delta_{\mathcal{J}}) \leq 2\varepsilon$. Equations (80) and (81) then contradict one another. Hence $F_{\alpha}(B) < \infty$.

Finally, for $M = I + K$ from (74), take $M_n = I + P_n K P_n$ and set $R = M_n^{-1} M$, which is close to I . Composition with the finite-coordinate map M_n^{-1} gives $\nu_{\alpha, R} \sim \mu_{\alpha}$. The diagonal part $D_R = \text{diag } R$ is positive, boundedly invertible, and satisfies $D_R - I \in \mathcal{S}_2$, so the one-coordinate scale criterion gives $\nu_{\alpha, D_R} \sim \mu_{\alpha}$. Thus

$$D_R^{-1} R = I + B, \quad \text{diag } B = 0, \quad B \in \mathcal{S}_2, \quad \|B\|_{\text{op}} \leq 1/10, \quad \nu_{\alpha, I+B} \sim \mu_{\alpha}.$$

Finite-coordinate and bounded diagonal composition preserve ℓ^{α} rows and compatibility; hence the finite-cut argument gives $F_{\alpha}(B) < \infty$. Since K_n is finitely supported,

$$M = M_n D_R (I + B) = D_R + (K_n D_R + D_R B + K_n D_R B).$$

The parenthesized term has finite column cost by (71). Restoring the signed permutation gives

$$\sum_j \left(\sum_{i \neq j} |k_{ij}|^2 \right)^{\alpha/2} < \infty,$$

while $K \in \mathcal{S}_2$ gives $\sum_i |k_{ii}|^2 < \infty$. Together these are $\mathfrak{e}_\alpha(TQ^{-1}) < \infty$.

If $\nu_{\alpha,T} = \mu_\alpha$, comparison of the joint characteristic functions, first for T and then for its compatible inverse, makes T^* a surjective isometry of ℓ^α . For $0 < \alpha < 2$, equality in

$$2\|u\|_\alpha^\alpha + 2\|v\|_\alpha^\alpha - \|u+v\|_\alpha^\alpha - \|u-v\|_\alpha^\alpha \geq 0$$

characterizes disjoint support. Hence the images of the coordinate vectors have disjoint supports; surjectivity makes each support a singleton, and norm preservation makes its entry ± 1 . Thus $T \in \mathcal{P}$. The converse follows from independence and symmetry. $\square$

4.1 Anisotropic stable Hellinger coercivity and sharpness

The proof gives more than a summability criterion: it measures the local change of law. The diagonal and off-diagonal directions have different orders, so the natural distance to the coordinate symmetries uses both terms of $\mathfrak{e}_\alpha$. For a finite matrix A , put

$$\mathfrak{e}_{\alpha,n}(A) = \min_{Q \in \mathcal{P}_n} \mathfrak{e}_\alpha(AQ^{-1}).$$

Proposition 4.2 (Anisotropic stable Hellinger coercivity). *Let $A \in \text{GL}(n, \mathbb{R})$. If all singular values of A lie in a fixed interval $[a, b] \subset (0, \infty)$, then there are positive constants depending only on α, a, b , and not on the dimension, such that*

$$c_{\alpha,a,b} \min\{\mathfrak{e}_{\alpha,n}(A), 1\} \leq h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \leq C_{\alpha,a,b} \min\{\mathfrak{e}_{\alpha,n}(A), 1\}. \quad (83)$$

Proof. The upper bound is (72), after removing the minimizing signed permutation. For the even convolution power k above, Fourier transformation and modulus contraction, followed by tensorization and pushforward under coordinatewise frequency addition, give

$$h^2(((A^{-1})^*)_{\#}((\rho_\alpha^\sharp)^{*k})^{\otimes n}, ((\rho_\alpha^\sharp)^{*k})^{\otimes n}) \leq k h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}). \quad (84)$$

Apply finite-variance coercivity to the rescaled $(\rho_\alpha^\sharp)^{*k}$ product. The singular values of $(A^{-1})^*$ lie in $[b^{-1}, a^{-1}]$, and

$$A - Q = -A((A^{-1})^* - Q)^*Q$$

compares the two distances up to a factor depending on a, b . Hence

$$h^2(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_{\alpha,a,b} \min \left\{ \min_{Q \in \mathcal{P}_n} \|A - Q\|_F^2, 1 \right\}. \quad (85)$$

Near a signed permutation, remove that permutation and write

$$d_{\text{diag}} = \sum_i |a_{ii} - 1|^2, \quad F = \sum_j \left(\sum_{i \neq j} |a_{ij}|^2 \right)^{\alpha/2}.$$

For $d_{\text{diag}} + F$ small, the identity is the nearest signed permutation, so (85) controls d_{diag} . With $D = \text{diag } A$ and $D^{-1}A = I + B$, $\text{diag } B = 0$ and $F_\alpha(B) \asymp_\alpha F$. The off-diagonal column lower bound and the diagonal scale upper bound give

$$h(A_{\#}\mu_{\alpha,n}, \mu_{\alpha,n}) \geq c_\alpha \sqrt{F} - C_\alpha \sqrt{d_{\text{diag}}}.$$

Choose $\eta \leq (c_\alpha/(2C_\alpha))^2$. Then $d_{\text{diag}} \leq \eta F$ implies $C_\alpha \sqrt{d_{\text{diag}}} \leq (c_\alpha/2)\sqrt{F}$. When $d_{\text{diag}} \geq \eta F$, use (85). This proves the local lower bound.

If the global lower bound failed, there would be $\varepsilon > 0$, dimensions n_k , and matrices $A_k \in \text{GL}(n_k, \mathbb{R})$, with singular values in $[a, b]$, such that $\mathfrak{e}_{\alpha, n_k}(A_k) \geq \varepsilon$ and, after passing to a subsequence,

$$\sum_k h^2((A_k)_\# \mu_{\alpha, n_k}, \mu_{\alpha, n_k}) < \infty.$$

The block-diagonal sum is boundedly invertible; its row map and inverse act blockwise and are compatible. Kakutani makes its image product equivalent to μ_α , so the classification supplies a signed permutation with finite anisotropic cost. A permutation column crossing a block contributes at least a^α , so all sufficiently late blocks are preserved. The corresponding block restrictions then give $\sum_k \mathfrak{e}_{\alpha, n_k}(A_k) < \infty$, contradicting the lower bound ε . This proves the global lower bound. $\square$

The exponent can be read directly from a single shear $T_t(x, y) = (x + ty, y)$. Conditioning on y and applying Plancherel gives

$$\begin{aligned} h^2((T_t)_\# \mu_{\alpha, 2}, \mu_{\alpha, 2}) &= 2 \int_{\mathbb{R}} (1 - e^{-|t\xi|^\alpha}) |\widehat{g}_\alpha(\xi)|^2 d\xi \\ &\sim 2|t|^\alpha \int_{\mathbb{R}} |\xi|^\alpha |\widehat{g}_\alpha(\xi)|^2 d\xi, \quad t \rightarrow 0. \end{aligned} \tag{86}$$

Thus the α -th order response comes from an explicit affinity calculation. Diagonal scales cost t^2 ; a quadratic Fisher form on all matrix directions would miss this difference.

Repeating the shear on independent pairs turns the local exponent into an exact summability threshold. Let

$$T_a(x_k, y_k)_{k \geq 1} = (x_k + a_k y_k, y_k)_{k \geq 1}.$$

Then $T_a - I \in \mathcal{S}_2$ exactly when $a \in \ell^2$, while [theorem 4.1](#) gives $\nu_{\alpha, T_a} \sim \mu_\alpha$ exactly when $a \in \ell^\alpha$. Any $a \in \ell^2 \setminus \ell^\alpha$ therefore gives a Hilbert–Schmidt perturbation with singular image law, even though the stable coordinate density is everywhere positive and has finite location and scale Fisher information. Now place all the mixing coefficients in a single input column:

$$S_u = I + u \otimes e_1, \quad u_1 = 0, \quad u \in \ell^2.$$

The inverse is $I - u \otimes e_1$, every row is finite, and the entire off-diagonal cost is $\|u\|_2^\alpha$. Hence $\nu_{\alpha, S_u} \sim \mu_\alpha$, even for $u \notin \ell^\alpha$. The same square-summable coefficients can therefore give singularity when attached to independent sources and equivalence when attached to one shared source. The column geometry records this dependence.

5 Density Zeros and Support Rigidity

A smooth density can vanish outside an interval. Contract just one coordinate, and part of that interval becomes inaccessible to the image law. The change has rank one and can be arbitrarily small, yet equivalence is lost. This obstruction is already present in finite dimension; no accumulation of errors is needed.

We separate this support obstruction from the Hellinger cost. The local geometry still measures small matrix perturbations, and absolute continuity still forces Hilbert–Schmidt

proximity to a coordinate symmetry. Within that neighbourhood, the singular part is exactly the mass sent into zeros of the density. Requiring the same avoidance for the inverse gives equivalence. For bounded and one-sided laws, these support constraints determine the operator outright.

5.1 Separating the mixing cost from the support obstruction

The path argument differentiates the square-root density on the whole line. It continues to work when that function reaches zero continuously enough to remain Sobolev. We assume

$$\begin{aligned} f \geq 0, \quad \int f = 1, \quad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad f \text{ non-Gaussian.} \end{aligned} \tag{87}$$

The boundary is part of this regularity condition. For example, the uniform density has zero derivative inside its support, but its square root jumps at the endpoints. Such a jump contributes a first-order Hellinger loss under a small rotation, as we will see below. A Sobolev square root has no boundary jump, and its derivative vanishes almost everywhere on its zero set. The identity $sg = -2g'$ therefore survives on the whole line.

Proposition 5.1 (Local Fourier coercivity with zeros of the density). *Under (87), the matrix Fisher operator decomposition, finite-dimensional Hellinger path bound, and dimension-free local Frobenius coercivity in Section 2 remain valid. Write $\mu = \rho^{\otimes \mathbb{N}}$, $\rho(dx) = f(x)dx$. There are constants $c, C, \delta > 0$, with $\delta < 1$, such that*

$$c\|K\|_{\mathcal{S}_2} \leq h(\mu, \nu_{I+K}) \leq C\|K\|_{\mathcal{S}_2} \quad (\|K\|_{\mathcal{S}_2} \leq \delta). \tag{88}$$

Moreover, for every $T \in \text{GL}(H)$, the following necessary condition survives:

$$\nu_T \ll \mu \implies T - Q \in \mathcal{S}_2 \text{ for some } Q \in \mathcal{P}. \tag{89}$$

For orthogonal transformations one may omit $xg' \in L^2$, retaining non-Gaussianity, $g \in W^{1,2}(\mathbb{R})$, and normalization to a centered law with variance one.

Proof. Define $s = -2g'/g$ on $\{g > 0\}$ and zero elsewhere. Since Sobolev derivatives vanish almost everywhere on zero sets, $sg = -2g'$ globally, and all cutoff identities—including $\mathbb{E}s(X) = 0$, $\mathbb{E}[Xs(X)] = 1$, and the formulas for J, κ —remain valid. The cases $J = 1$ and $\kappa = 0$ would respectively give $g' = -xg/2$ and $2xg' + g = 0$; the latter has no nonzero square-integrable solution. Thus the finite-dimensional proofs apply.

For $K \in \mathcal{S}_2$ with $I + K$ invertible, put $T_n = I + P_n K P_n$. The Hellinger path bound and invariance give, for large m, n ,

$$h(\nu_{T_n}, \nu_{T_m}) \leq C\|T_n^{-1}\|_{\text{op}}\|T_n - T_m\|_{\mathcal{S}_2}.$$

The square-root densities are therefore Cauchy in L^2 relative to a common dominating measure. Their norm-one limit is identified with ν_{I+K} by finite-row convergence, proving (88). The dominating measure here includes the parts outside the original positivity set. Polar approximants and the skew-generator bound from proposition 3.5 give the orthogonal case.

In lemma 3.11, the tail-symmetry and covariance lemmas already require only $\nu_T \ll \mu$; the other inputs are moment-only rigidity, the measurable inverse action, and local coercivity for finite-rank tail perturbations. The estimate above supplies the last input, so the same tail-matching proof yields (89). $\square$

To identify absolute continuity, we now ask where the image mass goes. Finite-coordinate approximants have densities wherever the original product is positive, and Hellinger convergence preserves this absolute continuity on that set. All singular mass must therefore lie outside it. For equivalence, apply the same argument to the inverse.

Theorem 5.2 (The linear equivalence group with density zeros). *Assume (87) and put $\mathcal{U} = \{x \in \mathbb{R} : f(x) > 0\}$. For every $T \in \text{GL}(H)$,*

$$\nu_T \ll \mu \iff \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i. \end{cases} \quad (90)$$

Consequently,

$$\{T \in \text{GL}(H) : \nu_T \sim \mu\} = \left\{ T \in \text{GL}(H) : \begin{cases} T - Q \in \mathcal{S}_2(H) \text{ for some } Q \in \mathcal{P}, \\ \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0, \\ \mu \left\{ x : \sum_j (T^{-1})_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i \end{cases} \right\}.$$

It is a subgroup of $\text{GL}(H)$. For orthogonal members the criterion holds with $T^{-1} = T^*$ and without $xg' \in L^2$.

Whenever $T - Q \in \mathcal{S}_2(H)$ for some $Q \in \mathcal{P}$, the Lebesgue decomposition of ν_T relative to μ is its restriction to $\mathcal{U}^{\mathbb{N}}$ plus its restriction to the complement: the former is absolutely continuous and the latter is singular.

Proof. Suppose $T - Q \in \mathcal{S}_2$ and put $K = TQ^{-1} - I$. Since $\nu_Q = \mu$, $\nu_T = \nu_{I+K}$. For $T_n = I + P_n K P_n$, [proposition 5.1](#) gives $h(\nu_{T_n}, \nu_T) \rightarrow 0$, hence total-variation convergence. Each ν_{T_n} has a Lebesgue density in its first n coordinates and agrees with μ on the tail, so positivity of μ_n on $\mathcal{U}^n$ and Fubini give $\nu_{T_n}|_{\mathcal{U}^{\mathbb{N}}} \ll \mu$. Passing to the limit proves the same for ν_T , while $\nu_T|_{(\mathcal{U}^{\mathbb{N}})^c} \perp \mu$. Consequently

$$\nu_T \ll \mu \iff \nu_T(\mathcal{U}^{\mathbb{N}}) = 1 \iff \mu \left\{ x : \sum_j T_{ij} x_j \notin \mathcal{U} \right\} = 0 \text{ for every } i.$$

Together with (89), this proves (90).

If $\nu_T \sim \mu$, the measurable inverse action gives $\nu_{T^{-1}} \sim \mu$, so the forward and inverse zero-set row conditions are necessary. Conversely, $T^{-1} - Q^{-1} = -T^{-1}(T - Q)Q^{-1} \in \mathcal{S}_2$, and these conditions with (90) give $\nu_T, \nu_{T^{-1}} \ll \mu$. The inverse action then yields

$$\mu = (\Phi_T)_{\#} \nu_{T^{-1}} \ll (\Phi_T)_{\#} \mu = \nu_T,$$

and hence equivalence. Measurable composition gives $\nu_{ST} = (\Phi_S)_{\#} \nu_T \sim \nu_S \sim \mu$ and closure under inverses follows above, proving the group assertion. In the orthogonal case, use the approximants from [proposition 3.5](#), which require only location Fisher information. $\square$

The forward condition prevents the image from entering a forbidden set. The inverse condition prevents it from leaving behind a region of positive original mass. These are different requirements, as the rank-one contraction below illustrates. Since nonzero row sums have densities, changing f on a Lebesgue-null set changes neither condition.

5.2 A rotating square: overlap without equivalence

The difference between boundary loss and accumulated loss is visible in a square. A small rotation removes thin triangles from its support. With uniform density, their mass is proportional to the angle; with a density vanishing smoothly at the boundary, their effect can be smaller. Start with ρ uniform on $[-\sqrt{3}, \sqrt{3}]$, let $\mu_{u,2} = \rho^{\otimes 2}$, and consider $T = \bigoplus_k O_{\theta_k}$. Write $\delta_k = \min_{m \in \mathbb{Z}} |\theta_k - m\pi/2| \in [0, \pi/4]$. For one block the affinity is the relative area of intersection of a square and its rotation. Removing the four corner triangles gives, for $0 < \delta \leq \pi/4$,

$$\begin{aligned} A(\mu_{u,2}, (O_\delta)_{\#} \mu_{u,2}) &= 1 - \frac{(\cos \delta + \sin \delta - 1)^2}{2 \sin \delta \cos \delta}, \\ 1 - A(\mu_{u,2}, (O_\delta)_{\#} \mu_{u,2}) &= \frac{\delta}{2} + O(\delta^2). \end{aligned}$$

At $\delta = 0$, the affinity is one. Each nonzero δ leaves positive area on both sides of the support difference. Consequently [theorem 5.4](#) gives the complete trichotomy

$$\begin{array}{l|l} \delta_k = 0 \text{ for every } k & \nu_T = \mu \\ 0 < \sum_k \delta_k < \infty & \nu_T \not\leq \mu, \quad \nu_T \not\leq \mu, \quad \mu \not\leq \nu_T \\ \sum_k \delta_k = \infty & \nu_T \perp \mu. \end{array}$$

In particular, $\theta_k = 1/k$ gives $T - I \in \mathcal{S}_2$ but $\nu_T \perp \mu$. With $\theta_k = 1/k^2$, the laws instead have positive affinity and neither is absolutely continuous with respect to the other.

To see the effect of a smoother boundary, let f be the variance-one rescaling of $(1-x^2)^r \mathbf{1}_{\{|x|<1\}}$, $r > 1$, put $\mu_{f,2} = (f(x) dx)^{\otimes 2}$, and set $J = 4\|(\sqrt{f})'\|_2^2 > 1$. The skew-generator formula gives

$$1 - A(\mu_{f,2}, (O_\theta)_{\#} \mu_{f,2}) = \frac{J-1}{4} \theta^2 + o(\theta^2). \quad (91)$$

Only signed-permutation rotations preserve the square support, but every rotation retains positive overlap near the origin. Continuity and (91) therefore replace $\sum_k \delta_k$ by $\sum_k \delta_k^2$ in the trichotomy, while equivalence still requires every $\delta_k = 0$.

5.3 When the support determines the operator

A bounded coordinate cannot be mixed freely with many independent copies without enlarging its range. Conditioning on finitely many inputs makes this observation precise: the width of the resulting sum is the original width multiplied by the ℓ^1 -norm of the coefficients. If both the map and its inverse preserve the range, both matrices have row norms at most one. Their inverse identities then force a signed permutation. For a half-line, a negative coefficient is ruled out by the unbounded tail, and the same argument leaves only ordinary permutations.

Write $\rho = \text{Law}(X)$, $\rho^- = \text{Law}(-X)$, and $\mu = \rho^{\otimes \mathbb{N}}$, with $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$. The row-series action is that of [lemma 3.1](#).

Theorem 5.3 (Linear equivalence for bounded and half-line coordinate laws). *Let $T \in \text{GL}(H)$.*

1. *Suppose $a = \text{ess inf } X < 0 < b = \text{ess sup } X$ are finite. Then $\nu_T \sim \mu$ forces T to be a signed permutation. The complete equivalence group is as follows:*

relation between ρ^- and ρ	allowed coordinate signs
$\rho^- = \rho$	all signs
$\rho^- \sim \rho, \rho^- \neq \rho$	only finitely many negative signs
$\rho^- \not\sim \rho$	no negative signs.

In particular, $a + b \neq 0$ permits only ordinary permutations. The support need not fill an interval, and a density is not required.

2. Suppose $\text{ess inf } X = a > -\infty$ and $\text{ess sup } X = +\infty$. Then

$$\nu_T \sim \mu \iff T \text{ is an ordinary coordinate permutation.}$$

The same conclusion holds for a law bounded above and unbounded below.

In the second case every equivalence is exact invariance. Failure of any of these equivalence criteria does not, by itself, imply singularity.

Proof. Assume $\nu_T \sim \mu$, and set $S = T^{-1}$. The measurable inverse action also gives $\nu_S \sim \mu$. For every row v of T or S , the series $Z = \sum_j v_j X_j$ converges in L^2 , and

$$\mathbb{E}[Z \mid X_1, \dots, X_m] = \sum_{j \leq m} v_j X_j. \quad (92)$$

In the bounded case $Z \in [a, b]$ almost surely, so its conditional expectation does too. The essential width of the finite sum in (92) is $(b - a) \sum_{j \leq m} |v_j|$; hence every row of T and S has ℓ^1 -norm at most one.

Let $A_{ij} = |T_{ij}|$ and $B_{ij} = |S_{ij}|$. The ℓ^2 row/column Cauchy–Schwarz inequality makes the entries of TS and ST absolutely convergent, so $AB \geq I$ and $BA \geq I$ entrywise. Tonelli and the row bounds give

$$\sum_j (AB)_{ij} = \sum_k A_{ik} \sum_j B_{kj} \leq 1, \quad \sum_j (BA)_{ij} \leq 1.$$

Thus $AB = BA = I$. For each i , choose k with $A_{ik}B_{ki} > 0$. The off-diagonal entries of AB force row k of B to vanish outside column i , and those of BA force row i of A to vanish outside column k . This gives a bijective monomial pattern. Its weights and reciprocals are at most one by the row bounds, so every weight is one; hence T is a signed permutation.

For a negative row sign, scalar equivalence requires $\rho^- \sim \rho$, and therefore $a = -b$. Each negative sign contributes $A(\rho, \rho^-)$ to the product affinity. This number equals one exactly when $\rho^- = \rho$; otherwise Kakutani's theorem [17] permits only finitely many negative signs. This proves the three alternatives.

In the lower-bounded, upper-unbounded case, centering and nondegeneracy give $a < 0$. Every finite sum in (92) is bounded below by a , so no coefficient can be negative. Its essential infimum is $a \sum_{j \leq m} v_j$, whence

$$a \sum_{j \leq m} v_j \geq a, \quad \text{hence} \quad \sum_j v_j \leq 1.$$

Thus T and S are nonnegative with row sums at most one, and the preceding inverse-matrix argument makes T an ordinary permutation. Such permutations preserve μ ; reflection handles the upper-bounded case. $\square$

If Y has a Gamma law with shape $r_\Gamma > 0$ and arbitrary positive scale, its standardization has endpoints $-\sqrt{r_\Gamma}$ and $+\infty$; hence only coordinate permutations are allowed for every positive shape and scale. For compact support, signs are governed by ρ versus ρ^- : if $\rho^- \sim \rho$ but $\rho^- \neq \rho$, only finitely many negative signs are allowed, whereas unequal endpoint magnitudes, as for a centered asymmetric Beta law, preclude them.

These support conclusions also cover densities with abrupt boundaries. For Gamma laws the zero-extended square root is Sobolev exactly when the shape exceeds two; for Beta laws both shapes must exceed two. The support argument treats all positive shapes in the same way.

What a one-coordinate contraction leaves behind. We return to the contraction that opened this section. It keeps every output in the original support, but it leaves a region of positive mass unreachable. This is exactly what the inverse row condition detects.

Let f be the rescaling to variance one of $C(1 - x^2)^2 \mathbf{1}_{\{|x| < 1\}}$, and write its support as $[-a, a]$. Set $\rho(dx) = f(x) dx$. Its zero-extended square root belongs to $W^{1,2}(\mathbb{R})$, and $x(\sqrt{f})' \in L^2(\mathbb{R})$. For $0 < c < 1$, set

$$T = c \oplus I, \quad f_c(y) = c^{-1} f(y/c), \quad \mu = \rho^{\otimes \mathbb{N}}.$$

Then $T - I$ has rank one and

$$\nu_T = (f_c(y) dy) \otimes \rho^{\otimes \{2,3,\dots\}} \ll \mu, \quad \mu \not\ll \nu_T.$$

Indeed, f_c is positive on $(-ca, ca)$ and zero on the rest of $(-a, a)$. Nevertheless,

$$A(\mu, \nu_T) = \int \sqrt{f(y) f_c(y)} dy > 0, \quad \sup_n E_n(T) < \infty.$$

The affinity sees the overlap on $(-ca, ca)$. The inverse row condition sees the missing mass on the rest of $(-a, a)$. The smooth boundary makes the Hellinger cost small, but it cannot restore that mass.

5.4 Block and finite-marginal criteria

The rotating-square examples combine two effects: each block can lose support, and the affinities of infinitely many blocks can tend to zero. The following form of the product criterion keeps these effects separate.

Theorem 5.4 (Block products without equivalence of the factors). *Let $\mu = \bigotimes_{k \geq 1} \lambda_k$ and $\nu = \bigotimes_{k \geq 1} \eta_k$ be probability products over finite-dimensional coordinate blocks. Set*

$$r_k = A(\lambda_k, \eta_k), \quad L = \sum_k -\log r_k \in [0, \infty], \quad -\log 0 = +\infty.$$

Then

$$A(\mu, \nu) = \prod_k r_k, \quad \mu \perp \nu \iff L = \infty. \quad (93)$$

If $L < \infty$, then

$$\begin{aligned} \nu \ll \mu &\iff \eta_k \ll \lambda_k \text{ for every } k, \\ \mu \ll \nu &\iff \lambda_k \ll \eta_k \text{ for every } k. \end{aligned}$$

Thus positive affinity is compatible with equivalence, with exactly one of $\nu \ll \mu$ and $\mu \ll \nu$, or with neither. For a linear block $A_k \in \text{GL}(d_k)$ and $\lambda_k = \rho^{\otimes d_k}$, where $\rho(dx) = f(x) dx$, the comparison density is

$$|\det A_k|^{-1} \prod_{i=1}^{d_k} f((A_k^{-1}y)_i).$$

The local absolute-continuity conditions concern the positivity set of this density modulo null sets, not merely the closure of its support. If $L < \infty$, take $m_k = (\lambda_k + \eta_k)/2$ with densities p_k, q_k . Then the masses of the parts of ν absolutely continuous relative to μ , and of μ relative to ν , are respectively

$$\prod_k \eta_k(\{p_k > 0\}), \quad \prod_k \lambda_k(\{q_k > 0\}). \quad (94)$$

Proof. Affinity tensorizes on finite block prefixes, and projected affinities decrease to global affinity by the conditional-expectation argument in [proposition 5.6](#). This proves (93).

The local absolute-continuity conditions are necessary by projection. If $\eta_k \ll \lambda_k$ for every k and $\prod_k r_k > 0$, set $\ell_n = \prod_{k \leq n} d\eta_k/d\lambda_k$. Independence gives

$$\|\sqrt{\ell_m} - \sqrt{\ell_n}\|_{L^2(\mu)}^2 = 2 \left(1 - \prod_{n < k \leq m} r_k \right) \rightarrow 0.$$

Their L^2 limit has norm one, and its square defines a probability measure which cylinder tests identify with ν ; hence $\nu \ll \mu$. Interchanging μ and ν proves the reverse assertion. The local condition cannot be omitted [17, 32]. For the mass formula, restrict to $C = \prod_k \{p_k q_k > 0\}$. Cauchy–Schwarz and $\prod_k r_k > 0$ make the conditional affinities have positive product, so the conditional product laws are equivalent. Their restrictions to C are precisely the two absolutely continuous parts, with the masses in (94). $\square$

Gamma scale changes and the effect of centering. The half-line classification concerns linear maps of centered coordinates. A dilation about the endpoint of a Gamma support is an affine map after centering. Its compensating translation keeps the endpoint fixed, allowing the familiar family of scale changes to reappear. For the uncentered shape- r_Γ , scale-one Gamma law, direct integration gives

$$\int_0^\infty \sqrt{f(y)c^{-1}f(y/c)} dy = \left(\frac{2\sqrt{c}}{1+c} \right)^{r_\Gamma} = \cosh\left(\frac{\log c}{2}\right)^{-r_\Gamma}.$$

Since $\log \cosh(u/2) \sim u^2/8$, Kakutani gives

$$\bigotimes_k \text{Law}(c_k Y) \sim \bigotimes_k \text{Law}(Y) \iff \sum_k (\log c_k)^2 < \infty, \quad c_k > 0. \quad (95)$$

Equivalently, $\sum_k (c_k - 1)^2 < \infty$. For $X = (Y - r_\Gamma)/\sqrt{r_\Gamma}$, the dilation $Y \mapsto cY$ becomes

$$X \mapsto cX + \sqrt{r_\Gamma}(c - 1),$$

rather than the homogeneous map $X \mapsto cX$.

Corollary 5.5 (Coordinatewise affine equivalence for Gamma products). *Let μ be the product of the shape- r_Γ Gamma law centered with variance one. For $T \in \text{GL}(H)$ and $b \in \ell^2$, the law of $\Phi_T(x) + b$ under μ is equivalent to μ if and only if there are a coordinate permutation π and numbers $c_i > 0$ such that*

$$(Tx)_i = c_i x_{\pi(i)}, \quad b_i = \sqrt{r_\Gamma}(c_i - 1), \quad \sum_i (\log c_i)^2 < \infty.$$

Proof. Set $S = T^{-1}$, so $Sb \in \ell^2$. Cauchy–Schwarz gives $\Phi_S(z + b) = \Phi_S(z) + Sb$ wherever the row limits exist, so the inverse is $z \mapsto \Phi_S(z) - Sb$. Translation by $d \in \ell^2$ changes a covariance constant C_λ to at most $2(C_\lambda + \|d\|_2^2)$, and a bounded linear pushforward by R changes it to at most $C_\lambda \|R\|_{\text{op}}^2$. Thus every affine-orbit law satisfies the hypothesis of [lemma 3.1](#); that lemma and the translation identity give the map and its inverse a common full-measure carrier. Conditioning finite row sums as in the proof of [theorem 5.3](#) shows that both T and T^{-1} have nonnegative entries: a negative coefficient would create an unbounded lower tail that no fixed shift can remove. The inverse identities then make T monomial with positive weights c_i . Equality of the scalar lower endpoints forces $-\sqrt{r_\Gamma}c_i + b_i = -\sqrt{r_\Gamma}$. Equation (95) completes both directions. $\square$

Finite-marginal criterion. Affinity tells us whether two laws overlap. When supports differ, full equivalence requires more: neither direction of the likelihood ratio may lose mass as more coordinates are observed. Uniform integrability expresses this requirement, and the following tail formula checks both directions at once.

Proposition 5.6 (Finite-marginal tails characterize equivalence). *Let μ, ν be any probability measures on $\mathbb{R}^{\mathbb{N}}$, and let $\mu^{(n)}, \nu^{(n)}$ be their first n marginals. Choose densities p_n, q_n with respect to any common dominating measure m_n . For $M > 1$, define*

$$\text{Tail}_n(M) = \int_{\{q_n > Mp_n\}} q_n dm_n + \int_{\{p_n > Mq_n\}} p_n dm_n.$$

Then

$$\mu \sim \nu \iff \lim_{M \rightarrow \infty} \sup_n \text{Tail}_n(M) = 0. \quad (96)$$

Equivalently, every finite pair is equivalent and its forward and reverse likelihood-ratio martingales are uniformly integrable in their respective reference measures. Without any finite-marginal equivalence assumption,

$$A(\mu^{(n)}, \nu^{(n)}) \downarrow A(\mu, \nu), \quad \sup_n [-2 \log A(\mu^{(n)}, \nu^{(n)})] < \infty \iff \mu \not\sim \nu.$$

Proof. If the limit in (96) is zero, a set where $p_n = 0 < q_n$ must have zero $\nu^{(n)}$ -mass for every n ; the second integral gives the reverse, so $\mu^{(n)} \sim \nu^{(n)}$. With $\ell_n(x) = q_n(\pi_n x)/p_n(\pi_n x)$, the two summands in $\text{Tail}_n(M)$ are

$$\mathbb{E}_\mu[\ell_n \mathbf{1}_{\{\ell_n > M\}}], \quad \mathbb{E}_\nu[\ell_n^{-1} \mathbf{1}_{\{\ell_n^{-1} > M\}}].$$

Thus the limit is exactly uniform integrability of the two mean-one likelihood-ratio martingales. Their L^1 limits have mass one, and cylinder tests prove mutual absolute continuity. Conversely, under equivalence they are conditional expectations of the full likelihood ratio and its reciprocal, hence uniformly integrable.

For affinity, take $m = (\mu + \nu)/2$, write $u = d\mu/dm$, $v = d\nu/dm$, and let u_n, v_n be their conditional expectations on the first n coordinates. Since $u, v \leq 2$, bounded martingale convergence gives

$$A(\mu^{(n)}, \nu^{(n)}) = \int \sqrt{u_n v_n} dm \longrightarrow \int \sqrt{uv} dm.$$

Conditional Cauchy–Schwarz gives monotonicity, and the limit vanishes exactly for mutually singular measures. $\square$

An asymmetric example with no fourth moment. The finite-variance classification does not rely on moment expansions beyond order two. To see the range this permits, consider the density proportional to

$$(1 + y^2)^{-(\tau+1)/2} e^{\vartheta \arctan y}, \quad 2 < \tau \leq 4, \quad \vartheta \in \mathbb{R} \setminus \{0\}.$$

Its pre-normalization score is $((\tau + 1)y - \vartheta)/(1 + y^2)$; it and its product with y are bounded, and these bounds persist under normalization. The fourth moment is infinite, and the two tails have unequal constants. After centering and normalization, the main theorem therefore classifies every bounded invertible mixing of this product. The bounded Fourier tests still detect its coordinate structure, and the tail exchanges still force exactly the Hilbert–Schmidt amount of deviation from an ordinary permutation.

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